Adversarial Dilated Large-Kernel Attention Networks for Cross-Domain Few-Shot Hyperspectral Image Classification

Xuan Liu, Lei Tong, *Member, IEEE*, Jun Zhou, *Senior Member, IEEE*, and Chuangbai Xiao, *Member, IEEE*

***Abstract*— The labeling hyperspectral image (HSI) data are time-consuming and laborious, resulting in limited labeled samples for training deep learning-based classifiers. To address this challenge, we propose ADLKAN, a cross-domain few-shot learning (FSL) framework based on an adversarial dilated large-kernel attention network. First, the ADLKAN consists of an adversarial dilated LKA feature extractor, which can enhance the ability to capture sparse features. The feature extractor includes two cooperative spectral–spatial attention blocks with large-kernel convolution, each of which extracts spectral and spatial features separately. These features are then fused to capture local contextual information and long-range dependencies in the spatial and spectral domains with attention networks. Second, ADLKAN also designs a multiscale conditional domain discriminator to alleviate the weakness of FSL under domain shift. The detailed multiscale features are extracted by three subdiscriminators on different input scales. Finally, the adversarial training is then adopted to optimize the embedding feature extractor and strengthen its consistency in uncertain regions of samples from different domains. Experimental results on five public HSI datasets show that the proposed ADLKAN outperforms other state-of-the-art methods.**

***Index Terms*— Adversarial dilated large-kernel attention (LKA), few-shot learning (FSL), hyperspectral image (HSI) classification, multiscale domain discriminator.**

I. Introduction

HYPERSPECTRAL images (HSIs) offer significantly richer spectral information than traditional color images. Typically comprising hundreds of continuous spectral channels, HSI can precisely identify the chemical composition of various materials [1], therefore, is extensively utilized in remote sensing, including precision agriculture [2], remote target detection [3], mineral exploration [4], and environmental monitoring [5].

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In the early days, classic machine learning methods have been widely used in HSI classification, including k-nearest neighbor (KNN) [6], [7], support vector machines (SVMs) [8], [9], [10], Bayesian classifier [11], decision tree [12], and logistic regression [13]. Traditional methods usually require manual feature extraction and use spectral and spatial information to determine the category of each pixel, achieving good results. However, these methods lack adaptability and generalization to large datasets and cannot achieve high-classification accuracy on complex scenarios.

Over the past several years, deep learning methods have been widely studied and developed. Deep learning methods have been applied to HSI classification and achieved encouraging results, improving the classification accuracy of ground objects [14]. Convolutional neural networks (CNNs), in particular, 2D-CNN [15] and 3D-CNN [16], demonstrated excellent capability in extracting hierarchical and deep spectral or joint spectral–spatial features, significantly improving the classification accuracy. More advanced CNN architectures, such as spectral–spatial residual network (SSRN) [17] and fast dense spectral space convolution (FDSSC) [18], have been introduced to alleviate the accuracy degradation or long training time problems, enabling more feasible classification solutions. Diffusion models [19], [20], [21], [22], [23] have also contributed to advancements in the domain of image classification. Nevertheless, CNN cannot extract long-range contextual information in sequential data or images. In addition, the convolutional kernels perform fixed local operations on the entire input data, making it difficult to flexibly adjust the degree of attention to different regions in the image.

In recent years, attention mechanisms (AMs) have been incorporated into HSI classification, offering a dynamic approach to adjusting the focus on different input regions through weighted distribution. This allows the model to selectively emphasize important information and directly attend to any position in the input data, unrestricted by distance. For example, considerations have been given to handle the sensitivity of 2-D and 3-D convolutions to HSI rotation [24], or extracting spectral–spatial information from the central pixel and similar surrounding pixels [25]. To capture the spectral–spatial cross attention, dual-branch structures [26], [27] have been proposed to capture fine-grained spatial information and extract spectral features, establishing interdependence between spectral and spatial domains. Despite

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their success, the self-AM overlooks the spectral dimension features, which are critical information in HSI [28]. The convolutional operations often suffer from an excessive number of weight parameters and lack global perceptibility. To address these challenges, large-kernel attention (LKA) networks have been proposed [28], [29] to simultaneously extract features in the spectral and spatial domains while using LKA to exploit both local and global features and capture long-range dependencies. However, LKA networks also face the challenge of overfitting due to sample scarcity.

This problem can be handled by cross-domain methods that leverage information from a source domain to enhance the classification performance in a target domain, so that more training data from another domain can be used. Cross-domain methods can be categorized into domain adaptation-based approaches [30], [31], [32] and fine-tuning-based approaches [33], [34], [35]. The domain adaptation focuses on feature-level adaptation, aiming to align the feature spaces of the source and target domains to minimize distribution differences. To this end, methods based on dictionary learning [30] and multikernel learning [36] are typical solutions. On the contrary, fine-tuning-based methods transfer the weights of the first layer of a pretrained network to the target model and then fine-tune the top layers using the backpropagation algorithm. However, both types of cross-domain methods have their limitations: domain adaptation methods struggle to classify new, unseen classes in the target domain, while fine-tuning methods are challenged by significant data bias between the source and target domains. Cross-domain few-shot learning (FSL) is a solution to this challenge. Approaches along this line [37], [38] integrate FSL with adversarial domain adaptation strategies for domain shift or refinement [39]. The domain discriminator employs adversarial training to guide the feature extractor in learning domain-invariant representations, narrowing the distribution gap between the source and target domains.

From the above, we can see that CNNs cannot capture long-distance dependencies, while AMs may neglect the spatial information. Moreover, a single-domain discriminator focuses on single-scale features, limiting its ability to model multiscale distribution differences between source and target domains. In addition, cross-domain FSL cannot avoid the domain shift problem caused by a single source of training data when performing in the source domain or target domain separately. To address these issues, we proposed ADLKAN, which uses a dilated large-kernel attention network to solve the problem of the lack of global perceptibility of convolutional operations and enhances the ability of large kernels to capture sparse features. It uses a multiscale alignment strategy to solve the limitations of a single discriminator in distinguishing spectrally similar categories. It employs an adversarial training strategy to optimize the embedded feature extractor, alleviating the domain shift problem caused by the limited diversity of data sources in single-domain FSL. Compared with SSLKA, it can extract sparse features and long-distance dependencies, while compared with ADAFSL, it can solve the domain transfer problem of single-domain FSL.

ADLKAN consists of an adversarial dilated LKA feature extractor and a multiscale conditional domain discriminator. First, two mapping layers ensure the same input dimension between the source and target domains. Then, a dilated LKA network is used to learn the spectral spatial metric space and perform FSL by comparing the distance between labeled and unlabeled data to address the sample scarcity problem. This attention network consists of two cooperative spectral–spatial attention blocks with large-kernel convolutions used to capture long-distance dependencies. In each block, we use a spectral attention module and a spatial attention module with large-convolutional kernels to extract features from the spectral and spatial domains separately and then fuse them to form the joint spectral–spatial features. Since the discriminative power of a single-condition domain discriminator is not high, we designed three subdiscriminators to take input vectors of three different scales, which are extracted through pooling operations, to enable higher discrimination capability with detailed multiscale features. Finally, we propose an adversarial training strategy for the embedded feature extractor. The domain shift is alleviated by strengthening the consistency of features in uncertain regions extracted from samples in different domains through adversarial learning.

The main contributions are summarized as follows.

1) A dilated LKA network is proposed for cross-domain scenarios. It consists of a spatial attention branch and a spectral attention branch, which are used to extract spectral–spatial features, and each branch uses the proposed LKA. In LKA, we use a nondilated small convolutional layer and multiple dilated small convolutional layers to enhance a nondilated large-kernel convolutional layer to obtain a nondilated convolutional layer with a larger sparse kernel. This design solves the problem of the lack of global perceptibility of the convolutional operation and enhances the ability of large kernels to capture sparse features.
2) To achieve accurate alignment, we propose a multiscale domain discriminator composed of three subdiscriminators operating at different scales to capture multiscale features. Each subdiscriminator independently extracts fine-grained features from the source and target domains across varying receptive fields and performs classification. The multiscale alignment strategy can effectively solve the limitations of single discriminators for distinguishing spectrally similar categories and has multiple positive effects, like enhancing local detail recognition, improving global feature understanding, and alleviating feature inconsistencies caused by domain shifts.
3) We propose an adversarial training strategy that operates exclusively on the source domain FSL and target domain FSL. In this approach, a domain discriminator assigns target domain labels to source domain samples and source domain labels to target domain samples. The resulting adversarial losses are calculated separately and are used to optimize the embedded feature extractor. This adversarial framework enhances the consistency of the embedded feature extractor when

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handling uncertain regions across domains, effectively mitigating the domain shift problem caused by the limited diversity of data sources in single-domain FSL.

The remainder of this article is structured as follows. Section II reviews related works. Section III describes the proposed ADLKAN framework. Section IV presents the experimental results and discusses the impact of various parameters on the ADLKAN design. Finally, Section V concludes this article.

II. Related Works

A. Cross-Domain FSL

Most FSL methods are essentially metalearning methods [40]. When only a small amount of labeled data is available in the target domain, the metaknowledge learned from the source class data can help classify the target class data. For a K-shot, C-way classification task, the goal of learning is to use the K-labeled training data of each category to learn a C-class classifier [41]. K usually takes a small value (1 or 5).

Under cross-domain FSL, two domain datasets are setup. The source domain dataset $\mathcal{D}_S$ with C_s class and the target domain dataset $\mathcal{D}_T$ with C_t class. C_s and C_t represent the number of classes in the source and target domains, respectively. To effectively learn metaknowledge, needs to be larger than C_t [42]. According to whether the data are labeled, the target domain dataset can be further divided into a few-sample dataset $\mathcal{D}_y$ with labeled data and a testing dataset $\mathcal{D}_n$ without labeled data, and $\mathcal{D}_y \cup \mathcal{D}_n$. During the training process, a small number of samples with episodes are used to simulate the FSL task. Each episode randomly selects samples from $\mathcal{D}_S$ or $\mathcal{D}_y$ to create specific source and target domain training tasks. Due to the scarcity of samples in $\mathcal{D}_y$, the data augmentation is usually performed by adding random Gaussian noise to the current known samples to increase training samples [43].

Take the source domain FSL as an example. First, C classes are randomly selected from C_s source classes. Then, K samples are selected from these C categories as the support set $\mathcal{S}_s = \{(\mathbf{x}_i, y_i)\}_{i=1}^{C\times K}$ and H samples are selected from the same category as the query set $\mathcal{Q}_s = \{(\mathbf{x}_j, y_j)\}_{j=1}^{C\times H}$. The class distribution of query sample $\mathbf{x}_j$ in the query set $\mathcal{Q}_s$ is

$$\mathbf{P}\left(y_j = k|\mathbf{x}_j \in \mathcal{Q}_s\right) = \frac{\exp\left(-d\left(f_\phi\left(\mathbf{x}_j\right), \mathbf{c}_k\right)\right)}{\sum_{k=1}^{C}\exp\left(-d\left(f_\phi\left(\mathbf{x}_j\right), \mathbf{c}_k\right)\right)} \quad (1)$$

where $d(\cdot,\cdot)$ represents the Euclidean metric, $\mathbf{c}_k$ is the embedded feature of the kth class in the support set, C is the number of categories in the source domain selected by each episode, f_ϕ represents the embedded feature extractor with parameter ϕ, $\mathbf{x}_j$ is the query set sample, and y_j is the label of $\mathbf{x}_j$.

In the training phase, the support set is used to train the embedding feature extractor, and the query set is used to calculate the distance to optimize the embedding feature extractor. The FSL training process in the source domain is similar to the FSL training process in the target domain. Iterative training tasks in cross-domain scenarios are completed by alternately executing FSL in the source and target domains.

B. Attention Mechanism

The AM is proposed based on the human visual system. It allocates more attention to key areas to obtain additional detailed information while suppressing irrelevant information. It plays an important role in target detection [44], [45], [46], [47], band selection [48], [49], [50], [51], and image classification [52], [53], [54], [55]. The channel AM [56] compresses spatial information on the feature map of each channel to generate a weight that represents the importance of the channel. Each channel weight is then multiplied by the feature map to obtain a weighted feature map for that channel. The spatial AM [57] dynamically adjusts the weight of each pixel based on its importance in a specific task, which allows the network to enhance image regions that contribute significantly to the task. The self-AM [58] is derived from NLP tasks and better captures the correlation between different elements in the data. In image processing, the self-AM solves the problem of long-distance dependencies in data by calculating the interactions between image patches.

In this article, we design a cooperative spectral–spatial attention module with large-convolutional kernels based on the AM. The local context information and long-range dependencies in the spatial and spectral domains can be fully captured through large-kernel convolution and AMs.

III. Methodology

In this section, we describe the details of the proposed ADLKAN framework, the embedded feature extractor based on the LKA mechanism, the multiscale conditional domain discriminator, and the adversarial training strategy of the embedded feature extractor.

A. ADLKAN Framework

The framework of ADLKAN in the training phase is shown in Fig. 1. It has two mapping layers, an embedding feature extractor with an LKA mechanism, and a multiscale conditional domain discriminator. ADLKAN consists of source domain FSL and target domain FSL, which are performed alternately during training. Take the source domain FSL as an example. First, the mapping layer processes the source domain and target data to the same dimension. Second, the embedding feature extractor embeds the source HSI and target HSI cubes into the spectral–spatial embedding space. In this space, samples of the same category shall be close to each other, and samples of different categories shall be far apart. FSL is performed by calculating the distance between each labeled sample class and unlabeled samples in the spectral–spatial embedding space of the source domain data. Then, the conditional domain discriminator alleviates the domain shift between the two domains and achieves domain distribution alignment. Finally, the embedding feature extractor is optimized through the adversarial training strategy. Specifically, the binary cross-entropy loss is calculated using the spectral–spatial embedding features of the target domain as the adversarial training loss of the source domain FSL. In the same way, we can perform the source domain FSL.

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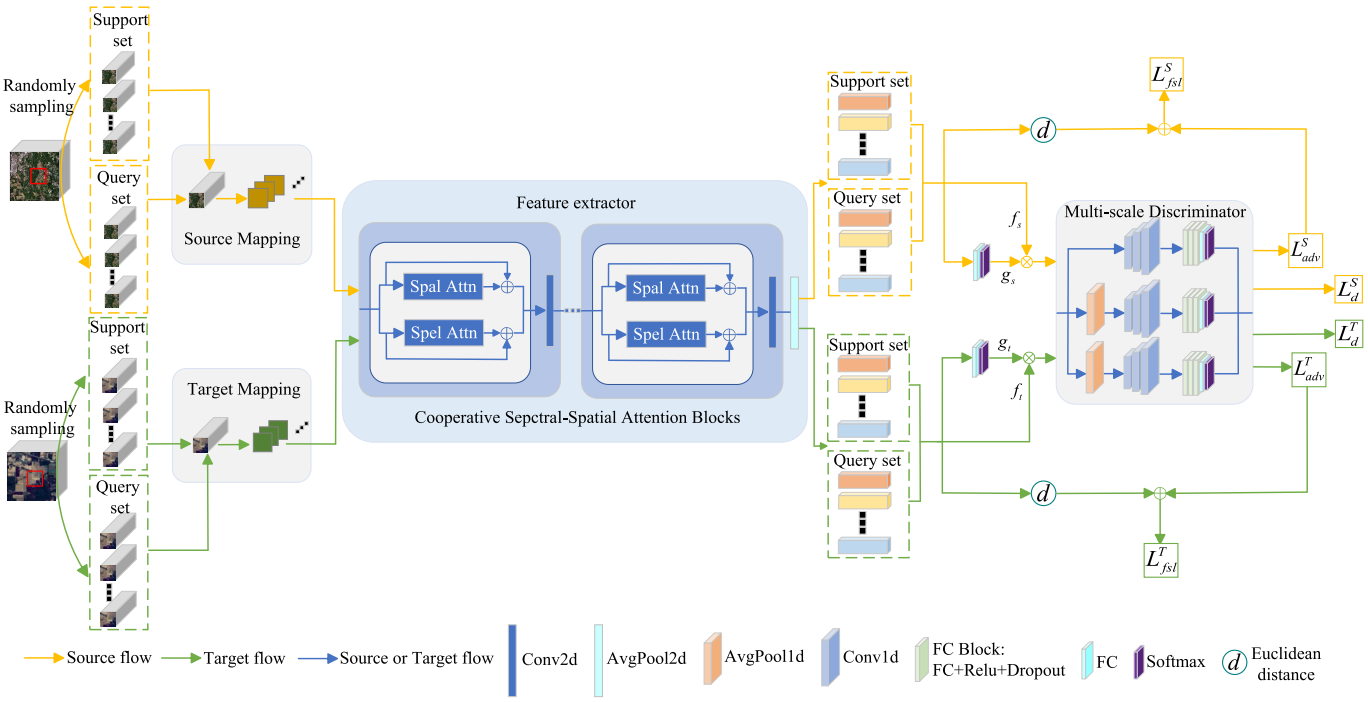


Fig. 1. Framework of the proposed ADLKAN in the training phase.

Therefore, the total loss function of the source domain FSL with adversarial domain adaptation is given by

$$L^s_{\text{total}} = L^s_{\text{fsl}} + \lambda L^s_{\text{adv}} + L^s_d \tag{2}$$

where λ is the balancing coefficient, L^s_d is the conditional domain adversarial loss of source domain FSL, and L^s_{adv} is the adversarial loss of source domain FSL optimized embedding feature extractor. According to the negative log probability of the query sample $\mathbf{x}$ and its true class label k, the classification loss of all query samples in the source domain can be expressed as follows:

$$L^s_{\text{fsl}} = E_{\mathcal{S}_s,\mathcal{Q}_s} \sum_{(\mathbf{x},y)\in\mathcal{Q}_s} -\log(\mathbf{p}(y=k|\mathbf{x})) \tag{3}$$

where $\mathcal{S}_s$ and $\mathcal{Q}_s$ are the support and query set from the source domain dataset, respectively.

Similarly, we can obtain the total loss function of the target domain FSL with adversarial domain adaptation

$$L^t_{\text{total}} = L^t_{\text{fsl}} + \lambda L^t_{\text{adv}} + L^t_d \tag{4}$$

where L^t_d is the conditional domain adversarial loss of the target domain FSL. L^t_{adv} is the adversarial loss of the target domain FSL optimized embedding feature extractor. The classification loss of all query samples in the target domain can be expressed as follows:

$$L^t_{\text{fsl}} = E_{\mathcal{S}_t,\mathcal{Q}_t} \sum_{(\mathbf{x},y)\in\mathcal{Q}_t} -\log(\mathbf{p}(y=k|\mathbf{x})) \tag{5}$$

where $\mathcal{S}_t$ and $\mathcal{Q}_t$ are the support and query set from the target domain dataset, respectively.

Finally, the total loss function for the ADLKAN is given by

$$L_{\text{total}} = L^s_{\text{total}} + L^t_{\text{total}}. \tag{6}$$

In addition, in the testing phase, the classification of unlabeled samples mainly includes three parts. First, the target mapping layer is used to reduce the sample dimension. Second, the spectral–spatial embedding features are extracted through the trained cooperative spectral–spatial attention network with large-convolutional kernels. Finally, the unlabeled samples are classified by the KNN classifier, and the classification map with labeled samples is generated.

B. Spectral–Spatial LKA Network

The embedding feature extractor consists of multiple cooperative spectral–spatial attention blocks (as shown in Fig. 2). Its network parameters are shown in Fig. 3. The cooperative spectral–spatial attention block consists of a spatial and a spectral attention branch, each containing a large-convolutional kernel attention block.

1) LKA Block: In our method, an attention module with large-convolutional kernels is applied to extract spectral–spatial features in HSI. The dilated reparam block uses multiple dilated small kernels to reparameterize the large kernel. Its hyperparameters include the size of the large kernel K, the size of the parallel convolutional layer k, and the dilation rate r. From a parameter point of view, such a dilated layer is equivalent to a nondilated convolutional layer with a larger sparse kernel, so the entire block can be equivalently converted into a single-large-kernel convolution. The dilated layer kernel of size k can obtain the nondilated layer kernel of size $(k-1)r+1$ by inserting zero entries. The case with five parallel layers shown in Fig. 4 is represented by $K = 13$, r = (1, 2, 3, 4, and 5), and k = (5, 7, 3, 3, and 3). The LKA module can be written as follows:

$$\mathbf{F_{lka}} = \sum_{r\in R,k\in K} \text{BN}\left(\text{Conv}^r_k(\mathbf{F})\right) \otimes \mathbf{F} \tag{7}$$

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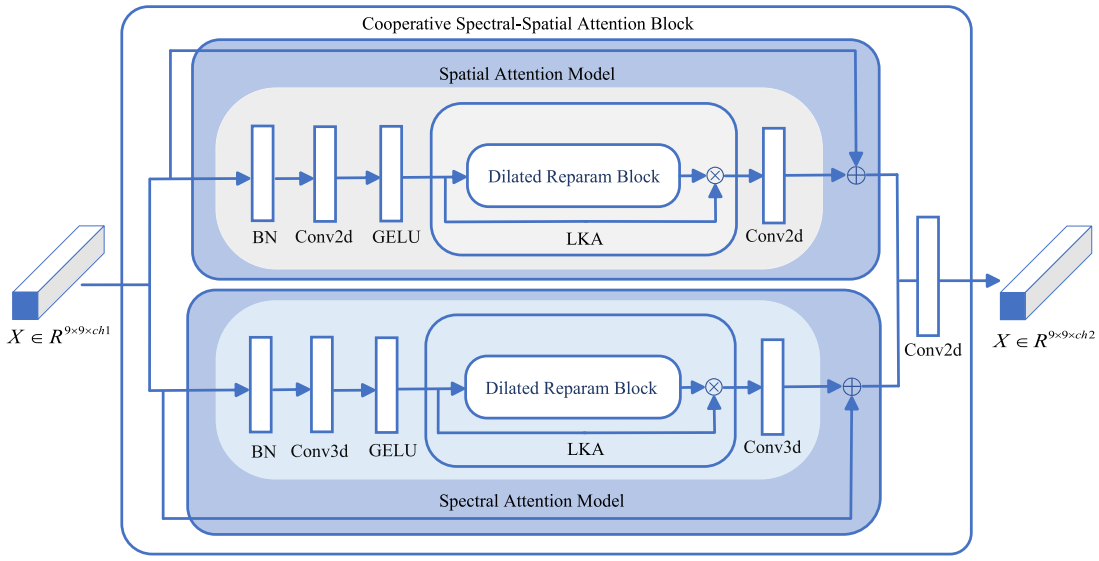


Fig. 2. Cooperative spectral–spatial attention block framework consists of parallel spectral attention modules and spatial attention modules. It fuses the spectral and spatial features through 2-D convolution.

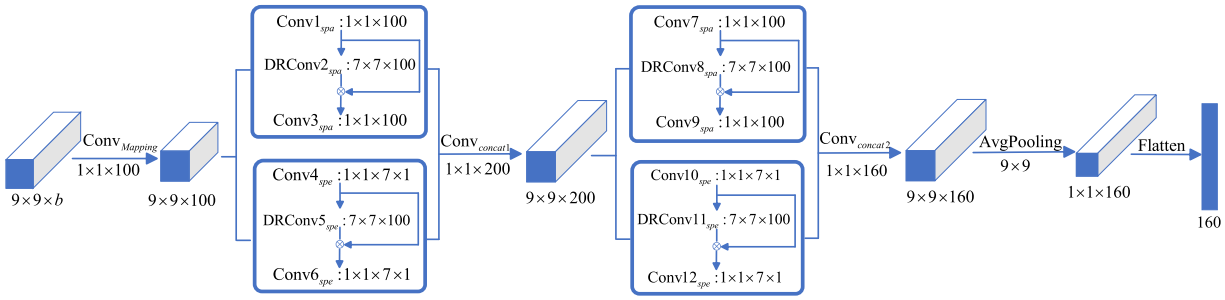


Fig. 3. Parameters of the proposed mapping layer and embedding feature extractor.

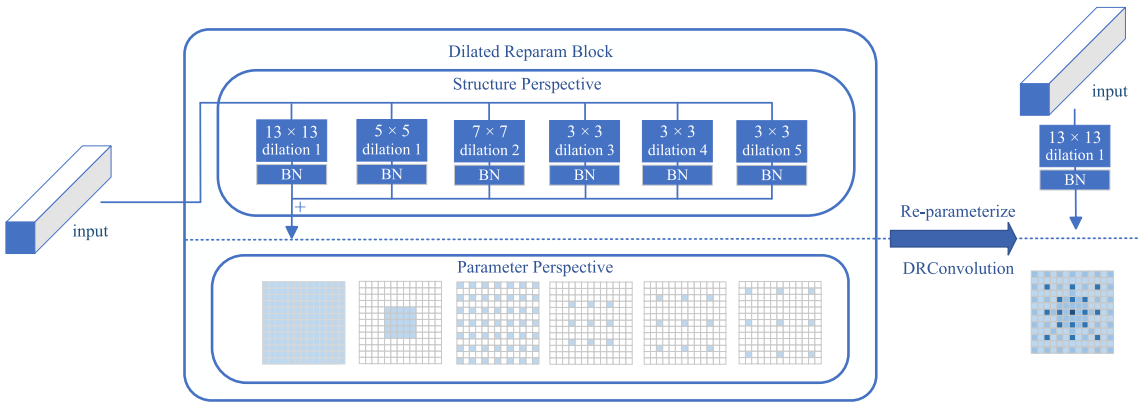


Fig. 4. Dilated reparam block enhances the nondilated large kernel by using multiple dilated small kernels. From a parameter perspective, these dilated small kernels are equivalent to a large-nondilated sparse kernel, so the entire block can be equivalently converted into a single-large-kernel convolution. For example, with a kernel size of $K = 13$, we can get a larger K by using more dilated layers.

where Conv_k^r represents a convolutional kernel with size k and dilation rate r, R and K represent the dilation rate set and the convolutional kernel size set, respectively, and $\otimes$ represents the element-by-element multiplication. $\mathbf{F}$ and $\mathbf{F_{lka}}$ are the input data and output data, respectively.

2) Cooperative Spectral–Spatial Attention Block: As shown in Fig. 2, the upper part of the structure represents the spatial attention branch, and the lower part represents the spectral attention branch. In the spatial attention branch, the input features go through a batch normalization (BN) layer, a 1×1 2-D convolutional layer, and a Gaussian error linear unit (GELU) to enhance the nonlinear representation. Then, the LKA module composed of the dilated reparam block, generates a spatial attention map with local and global spatial features. Finally,

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a 2-D convolution is also used to extract spatial features. The spatial feature $\mathbf{F_{spa}}$ can be expressed as follows:

$$\mathbf{F_{spa}} = \Gamma(\mathbf{F_{lka}}(\mathrm{GELU}(\Gamma(\mathrm{BN}(\mathbf{F_m}))))) \oplus \mathbf{F_m} \tag{8}$$

where $\mathbf{F_m}$ represents the preprocessed data, Γ represents a 2-D convolution with a kernel size of 1×1, and $\oplus$ represents the elementwise summation.

The input features of the spectral attention branch use a 3-D convolutional kernel size $1 \times 1 \times 7$ to extract the spectral information after passing through the BN layer, then passing through the GELU layer and LKA module to generate a weighted feature map. At the end of the spectral attention branch, 3-D convolution is used again to extract weighted spectral information. The spectral feature $\mathbf{F_{spe}}$ can be expressed as follows:

$$\mathbf{F_{spe}} = (\mathbf{F_{lka}}(\mathrm{GELU}(\Upsilon(\mathrm{BN}(\mathbf{F_m}))))) \oplus \mathbf{F_m} \tag{9}$$

where Υ represents a 3-D convolution with a kernel size of $1 \times 1 \times 7$.

Finally, the extracted spatial and spectral attention features are combined through concatenation to achieve feature fusion, as described in (10). A 2-D convolution with a kernel size of 1×1 is used to merge the spatial features $\mathbf{F_{spa}}$ and the spectral features $\mathbf{F_{spe}}$

$$\mathbf{F_{fus}} = \mathrm{GELU}\big(\Gamma\big(\mathrm{Concat}\big(\mathbf{F_{spe}}, \mathbf{F_{spa}}\big)\big)\big) \tag{10}$$

where $\mathbf{F_{fus}}$ is the final output of the feature fusion stage and Concat means concatenating the features in the spatial and spectral domains.

C. Multiscale Conditional Domain Discriminator

The conditional domain discriminator $\mathbf{D}$ solves the domain shift problem between the source distribution $\mathbf{P}_s(\mathbf{x})$ and the target distribution $\mathbf{P}_t(\mathbf{x})$ in the ADLKAN framework. The classifier predicts $\mathbf{g}$. It contains discriminative information and can condition the adversarial adaptation of the spectral–spatial feature $\mathbf{f}$. The domain adversarial loss function L is defined on the domain discriminator $\mathbf{D}$, the feature representation $\mathbf{f} = \mathbf{F}(\mathbf{x})$, and the classifier prediction $\mathbf{g} = \mathbf{G}(\mathbf{x})$. However, a single-conditional domain discriminator cannot identify more detailed features. It shows wrong discriminative decisions when facing similar discriminative information in different domains, therefore, is easily deceived and cannot effectively alleviate the domain shift problem.

Inspired by [37] and [59], we propose a multiscale conditional domain discriminator as shown in the right part of Fig. 1. First, three input vectors of different scales are obtained through the pooling layer. Then, three subdiscriminators are run on vectors of various scales, and their average adversarial loss is calculated as the final loss. Each subdiscriminator contains a 1-D convolutional layer and a multilayer perceptron consisting of five fully connected layers. Except for the last layer, each fully connected layer is followed by a rectified linear unit (ReLU) nonlinear activation function and a dropout layer. The convolutional layer extracts detailed features of inputs of different scales, and the multilayer perceptron is used to complete the final discrimination.

Let $\mathbf{h} = (\mathbf{f}, \mathbf{g})$ be the joint variable of $\mathbf{f}$ and $\mathbf{g}$. Use a multilinear map $\mathbf{f} \otimes \mathbf{g}$ (defined as the outer product of multiple random vectors) to capture the multimodal structure behind the complex data distribution fully. However, multilinear mapping faces the problem of dimensionality explosion. To solve the dimensionality explosion problem, ADLKAN uses random multilinear mapping to replace multilinear mapping, which can be expressed as follows:

$$\mathbf{T}(\mathbf{h}) = \begin{cases} \mathbf{f} \otimes \mathbf{g}, & d_f \times d_g \leq 1024 \\ \left(1/\sqrt{d}\right)\left(R_f \mathbf{f}\right) \odot \left(R_g \mathbf{g}\right), & \text{otherwise} \end{cases} \tag{11}$$

where $\odot$ denotes the elementwise product, d_f and d_g denote the dimensions of $\mathbf{f}$ and $\mathbf{g}$, $R_f \in R^{d \times d_f}$ and $R_g \in R^{d \times d_g}$ denote two random matrices, which are sampled only once and fixed in the training phase, and $d \ll d_f \times d_g$. If the dimension of the multilinear map is greater than 1024, we will use a random multilinear map. The conditional domain-adversarial loss can be expressed as follows:

$$L_d = \min_{\mathbf{D}} \max_{\mathbf{T}} = \frac{1}{N}\left(-E_{\mathbf{x}_i^s \sim \mathbf{P}_s(\mathbf{x})} \sum_{n=1}^{N} \log\left[D_n\left(\mathbf{T}\left(\mathbf{h}_i^s\right)\right)\right] - E_{\mathbf{x}_j^t \sim \mathbf{P}_t(\mathbf{x})} \sum_{n=1}^{N} \log\left[1 - D_n\left(\mathbf{T}\left(\mathbf{h}_j^t\right)\right)\right]\right) \tag{12}$$

where $D()$ is the probability that the discriminator predicts that $\mathbf{x}$ is a source domain sample, $1 - D()$ is the probability that the discriminator predicts that $\mathbf{x}$ is a target domain sample, and N is the number of subdiscriminators. The discriminator $\mathbf{D}$ minimizes the above function, while the feature extractor $\mathbf{F}$ and classifier $\mathbf{G}$ maximize it.

D. Adversarial Training of Embedding Feature Extractors

In the ADLKAN framework, source and target domain FSL are performed alternately. In each FSL execution process, only the HSI information of one domain are used to train the embedding feature extractor. Taking source domain FSL as an example, the embedded feature extractor trained with source domain HSI shows poor performance when extracting features from target domain HSI, affecting the final classification accuracy. The reason is that there is a domain shift between different domains, which makes the feature extractor unable to recognize uncertain features of different domains. Although alternating FSL and conditional domain adversarial strategy can alleviate domain shift in the entire training process, the domain shift problem caused by the single source of training data cannot be avoided when each FSL is executed separately.

To solve this problem, we use an adversarial training strategy to optimize the embedding feature extractor, as shown in the right part of Fig. 1. In the source domain FSL, the target domain data are discriminated as the corresponding label of the source domain through the domain discriminator to calculate the adversarial loss L_{adv}^s. The adversarial loss of the source domain FSL can be expressed as follows:

$$L_{\mathrm{adv}}^s = \min_{\mathbf{D}} \max_{\mathbf{T}} = -E_{\mathbf{x}_i^s \sim \mathbf{P}_s(\mathbf{x})} \frac{1}{N} \sum_{n=1}^{N} \log\left[1 - D_n\left(\mathbf{T}\left(\mathbf{h}_i^s\right)\right)\right]. \tag{13}$$

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Similarly, the adversarial loss of the target domain FSL can be expressed as follows:

$$L_{\text{adv}}^{t} = \min_{\mathbf{D}} \max_{\mathbf{T}} = -E_{\mathbf{x}_j^t \sim \mathbf{P}_t(\mathbf{x})} \frac{1}{N} \sum_{n=1}^{N} \log\left[D_n\left(\mathbf{T}\left(\mathbf{h}_j^t\right)\right)\right]. \quad (14)$$

By optimizing the embedding feature extractor using an adversarial training strategy, its consistency in uncertain features in different domains can be enhanced. This method can alleviate the domain offset problem caused by the singleness of the data source when executing a single FSL. The detailed description of the ADLKAN training process is presented in Algorithm 1.

Algorithm 1 Pseudocode for the Training Process of the Proposed ADLKAN

Input: $\mathcal{S}_t$, $\mathcal{Q}_t$ of $\mathcal{D}_y$, $\mathcal{S}_s$, $\mathcal{Q}_s$ of $\mathcal{D}_S$, $\mathcal{D}_n$; the number of training episodes; learning rate η; kernel size k; balance item λ.

Output: The classification accuracy of each class of the target dataset.

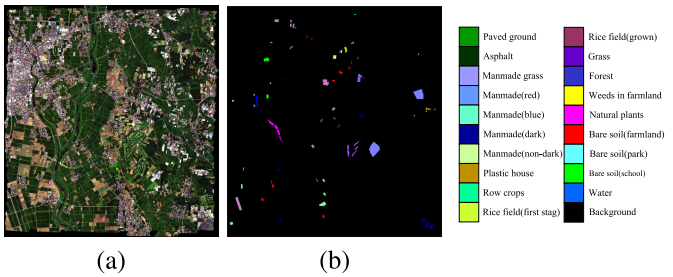
Training Stage
for episode=1 to episodes **do**
  if episode mode 2 == 0 **then**
    Randomly selected $\mathcal{S}_s$, $\mathcal{Q}_s$ from $\mathcal{D}_S$;
    Extract $\mathbf{f}_s$ through the network described in Fig. 3.;
    Calculate L_{fsl}^{s} by Eq.(3);
    Calculate $\mathbf{g}_s$ of $\mathcal{S}_s$ and $\mathcal{Q}_s$ through softmax
    Calculate joint features $\mathbf{T}(\mathbf{h}(\mathbf{f}_s, \mathbf{g}_s))$ by Eq.(11);
    Calculate L_d^s and L_{adv}^{s} by Eqs.(12) and (13);
    $L_{\text{total}}^{s} = L_{\text{fsl}}^{s} + \lambda L_{\text{adv}}^{s} + L_d^s$
    L_{total}^{s}.backward()
  else
    Randomly selected $\mathcal{S}_t$, $\mathcal{Q}_t$ from $\mathcal{D}_y$;
    Extract $\mathbf{f}_t$ through the network described in Fig. 3.;
    Calculate L_{fsl}^{t} by Eq.(5);
    Calculate $\mathbf{g}_t$ of $\mathcal{S}_t$ and $\mathcal{Q}_t$ through softmax
    Calculate joint features $\mathbf{T}(\mathbf{h}(\mathbf{f}_t, \mathbf{g}_t))$ by Eq.(11);
    Calculate L_d^t and L_{adv}^{t} by Eqs.(12) and (14);
    $L_{\text{total}}^{t} = L_{\text{fsl}}^{t} + \lambda L_{\text{adv}}^{t} + L_d^s$
    L_{total}^{s}.backward()
  end if
  Testing Stage
  if episode mode 500 == 0 **then**
    Reduce the $\mathcal{D}_n$ dimension by target mapping;
    Extract $\mathbf{f}_n$ through the network described in Table I;
    Classify $\mathbf{f}_n$ by KNN classifier;
    Calculate the accuracy of each class of $\mathcal{D}_n$
  end if
end for
$L_{\text{total}} = L_{\text{total}}^{s} + L_{\text{total}}^{t}$
return Outputs

IV. Experiments

To verify the effectiveness of the proposed ADLKAN method, we conducted experiments using the Chikusei dataset as the source domain and the University of Pavia (UP), Pavia Center(PC), Salinas, Kennedy Space Center (KSC), FL, USA, and Indian Pines (IP) datasets as the target domain.

TABLE I
Land Cover Classes and Numbers of Samples in the Chikusei Dataset

Class	Name	Samples
1	Water	2845
2	Bare soil(school)	2859
3	Bare soil(park)	286
4	Bare soil(farmland)	48525
5	Natural plants	4297
6	Weeds in farmland	1108
7	Forest	20516
8	Grass	6515
9	Rice field(grown)	13369
10	Rice field(first stage)	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade(non-dark)	1220
14	Manmade(dark)	7664
15	Manmade(blue)	431
16	Manmade(red)	222
17	Manmade grass	1040
18	Asphalt	801
19	Paved ground	145
-	Total	77592



(a) (b)

Fig. 5. Chikusei dataset. (a) False-color image. (b) Ground-truth map.

A. Experimental Dataset

1) Chikusei Dataset: The Chikusei dataset was collected using hyperspectral visible/near-infrared cameras (Hyperspec-VNIR-C) in Chikusei, Ibaraki, Japan, on July 29, 2014. This dataset contains 19 categories of 128 spectral bands with a spectral range of 363–1018 nm. Its spatial size is 2517×2335, with a spatial resolution of 2.5-m per pixel. The details of the Chikusei dataset are shown in Table I. The false-color image and corresponding ground-truth map are shown in Fig. 5.

2) IP Dataset: The IP dataset was collected with the AVIRIS sensors in Indiana in 1992. This dataset contains 16 ground categories and 200 bands ranging from 400 to 2500 nm. Its image size is 145×145, with a spatial resolution of 20-m per pixel. The details of the IP dataset are provided in Table II. The false color image and real image are shown in Fig. 6.

3) Salinas Dataset: The Salinas dataset was captured in CA, USA, using the AVIRIS sensors. It contains 16 crop categories, 204 available band data, a spatial resolution of 3.7-m per pixel, and a size of 512×217. The details of the Salinas dataset are described in Table III. The false color and real images are shown in Fig. 7.

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TABLE II
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN THE IP DATASET

Class	Name	Samples
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-tree	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Trees-Drives	386
16	Stone-Steel-Towers	93
-	Total	10249



(a) (b)

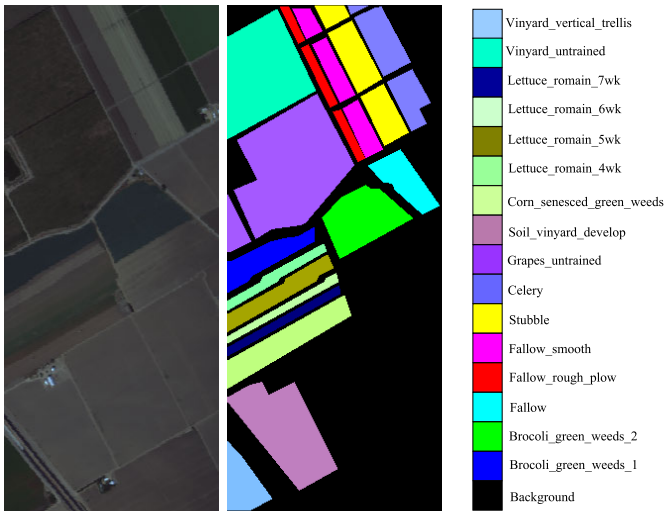
Fig. 6. IP dataset. (a) False-color image. (b) Ground-truth map.

TABLE III
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN THE SALINAS DATASET

Class	Name	Samples
1	Brocoli_green_weeds_1	2009
2	Brocoli_green_weeds_2	3726
3	Fallow	1976
4	Fallow_rough_plow	1394
5	Fallow_smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes_untrained	11271
9	Soil_vinyard_develop	6203
10	Corn_senesced_green_weeds	3278
11	Lettuce_romaine_4wk	1068
12	Lettuce_romaine_5wk	1927
13	Lettuce_romaine_6wk	916
14	Lettuce_romaine_7wk	1070
15	Vinyard_untrained	7268
16	Vinyard_vertical_trellis	1807
-	Total	54129

4) KSC Dataset: The KSC dataset was captured by the AVIRIS sensors in 1996 at the KSC. It covers 13 ground categories, including 176 bands, and the original image size is 512 × 614. The details of the KSC dataset are given in Table IV. The false color and real images are shown in Fig. 8.

5) UP Dataset: The UP dataset was collected by the Reflective Optical System Imaging Spectrometer (ROSIS) sensor in Pavia, Italy, in 2003. This dataset contains nine ground categories, 103 spectral bands in 430–860 nm. Its spatial size is 610 × 340, with a spatial resolution of 1.3-m per pixel. The details of the UP dataset are given in Table V. The false color and real images are shown in Fig. 9.

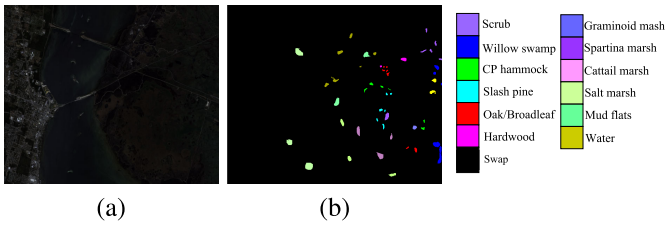


(a) (b)

Fig. 7. SALINAS dataset. (a) False-color image. (b) Ground-truth map.

TABLE IV
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN THE KSC DATASET

Class	Name	Samples
1	Scrub	761
2	Willow swamp	243
3	CP hammock	256
4	Slash pine	252
5	Oak/Broadleaf	161
6	Hardwood	229
7	Swamp	105
8	Graminoid mash	431
9	Spartina marsh	520
10	Cattail marsh	404
11	Salt marsh	419
12	Mud flats	503
13	Water	927
-	Total	5211



(a) (b)

Fig. 8. KSC dataset. (a) False-color image. (b) Ground-truth map.

6) PC Dataset: The PC dataset was acquired by the ROSIS sensor during an aerial survey over Pavia, a city in Northern Italy. Comprising 1906×715 pixels, it features a spatial resolution of 1.3-m per pixel. The dataset encompasses 102 spectral bands, spanning from 430 to 860 nm. The ground-truth data for this dataset includes nine distinct classes, which are characteristic of an urban area. The false color image and the corresponding ground-truth map are presented in Fig. 10. The samples of the PC dataset are detailed in Table VI.

B. Experimental Settings

All experiments were conducted on a desktop computer with an NVIDIA GeForce RTX 3090 GPU and implemented

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TABLE V
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN THE UP DATASET

Class	Name	Samples
1	Asphalt	6631
2	Meadows	18649
3	Cravel	2099
4	Trees	3064
5	Sheets	1345
6	Bare soil	5029
7	Bitumen	1330
8	Bricks	3682
9	Shadow	947
-	Total	42776

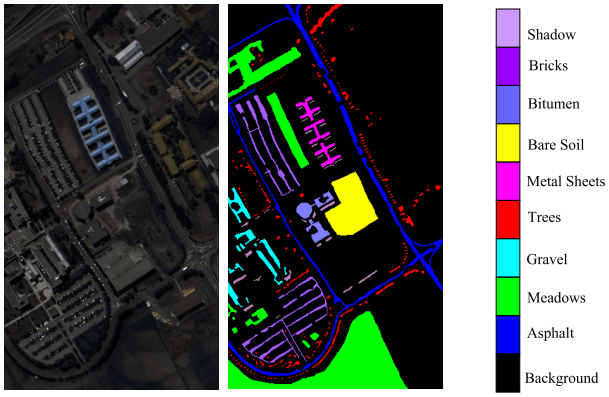


Fig. 9. UP dataset. (a) False-color image. (b) Ground-truth map.

TABLE VI
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN THE PC DATASET

Class	Name	Samples
1	Water	65971
2	Trees	7598
3	Meadows	3090
4	Bricks	2685
5	Bare Soil	6584
6	Asphalt	9248
7	Bitumen	7287
8	Tiles	42826
9	Shadows	2863
-	Total	148152

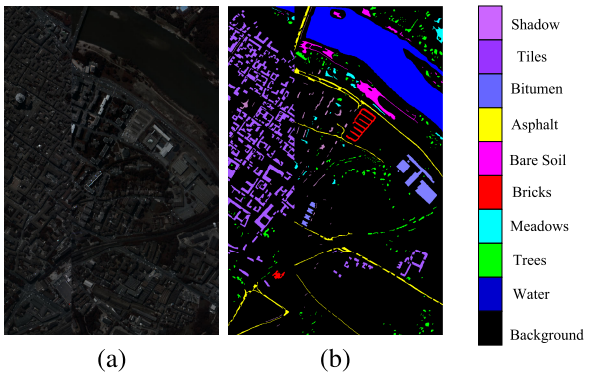


Fig. 10. PC dataset. (a) False-color image. (b) Ground-truth map.

in PyTorch on Ubuntu 18.04. Referring to the experimental settings in [60], the spatial size of the input cube was set as a

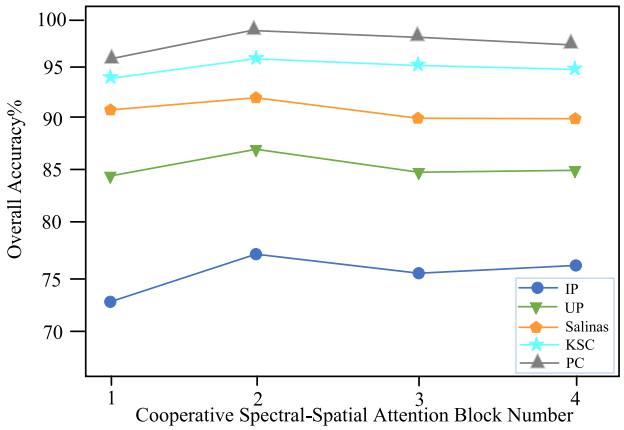


Fig. 11. OA with different numbers of attention blocks on each dataset.

9×9 neighborhood. ADLKAN was trained using the Adam optimizer, and all convolutional kernels were initialized using Xavier normalization. Training iterations were set to 10 000 to fully train the network. The learning rate was set to 0.0001, and the balance coefficient λ of adversarial training was set to 0.1. In the FSL training phase, the K-shot, C-way method was used for training, where K represents the number of samples selected for training in each set, and C represents the number of categories in the target dataset. In theory, the more samples in the support set and the fewer samples in the query set, the better it can simulate the small sample scenario in the target dataset. Therefore, we set the number of samples in the support and query sets to 1 and 19, respectively. In the experiments, we used overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa) to evaluate the classification performance of different methods. To avoid any bias caused by random sampling, all the experiments were conducted ten times with the same experimental settings.

C. Parameter Settings

To determine the impact of the number of cooperative spectral space attention blocks on the experimental results, we set the number of attention blocks to 1–4, respectively, for experiments. As shown in Fig. 11, when the number of cooperative spectral–spatial attention blocks is 2, the best classification results are achieved on OA. When the number of blocks is 1, the feature extractor cannot effectively extract features in both spatial and spectral domains, resulting in poor performance. Overfitting leads to low accuracy when the number of blocks is 3 or 4. Therefore, we set the number of layers of the collaborative spectral–spatial attention module to 2 in all experiments.

In ADLKAN, the balance coefficient λ of adversarial training enables a good compromise between adversarial training loss and other classification losses, so that the feature extractor can learn common features between domains, reduce the impact of differences between domains, and alleviate the domain shift problem caused by the diversity of data sources in single-domain FSL. To examine the impact of λ on the experimental results, we set it to 0.05, 0.1, 0.15, and 0.2,

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TABLE VII
EXPERIMENTAL RESULTS OF DIFFERENT λ SIZES ON UP, SALINAS, KSC, IP, AND PC DATASETS

Matrices	Datasets	λ=0.05	λ=0.1	λ=0.15	λ=0.2
OA%	IP	76.05	**77.70**	75.48	75.40
AA%		85.83	**86.56**	85.50	85.37
Kappa×100		73.06	**74.89**	72.45	72.34
OA%	SALINAS	91.36	**92.52**	91.98	91.85
AA%		95.27	**95.96**	95.48	95.50
Kappa×100		90.41	**91.68**	91.09	90.94
OA%	KSC	95.91	**96.58**	95.94	95.79
AA%		93.86	**95.04**	94.09	93.78
Kappa×100		95.44	**96.19**	95.48	95.31
OA%	UP	85.48	**86.64**	85.31	85.58
AA%		87.96	**88.87**	87.75	88.09
Kappa×100		81.36	**82.82**	81.17	81.53
OA%	PC	97.46	**97.75**	97.58	97.30
AA%		94.49	**94.99**	94.46	94.17
Kappa×100		96.41	**96.82**	96.59	96.19

TABLE VIII
EXPERIMENTAL RESULTS OF LKA WITH DIFFERENT CONVOLUTIONAL KERNEL SIZES ON UP, SALINAS, KSC, IP, AND PC DATASETS

Matrices	Datasets	7×7	9×9	11×11	13×13	15×15	17×17
OA%	IP	76.47	75.68	76.65	**77.70**	77.57	76.19
AA%		85.92	85.26	85.61	**86.56**	86.44	85.74
Kappa×100		73.49	72.59	73.67	**74.89**	74.76	73.18
Params(K)		**116.13**	128.93	144.93	164.13	186.04	212.13
FLOPs(M)		**151.21**	167.80	188.53	213.42	242.45	275.62
OA%	SALINAS	91.67	91.88	92.25	**92.52**	92.10	92.33
AA%		95.45	95.51	95.75	**95.96**	95.68	95.87
Kappa×100		90.75	90.97	91.39	**91.68**	91.23	91.48
Params(K)		**116.13**	128.93	144.93	164.13	186.53	212.13
FLOPs(M)		**151.21**	167.80	188.53	213.42	242.45	275.62
OA%	KSC	96.35	96.37	96.42	**96.58**	96.53	96.41
AA%		94.58	94.68	94.78	**95.04**	94.79	94.67
Kappa×100		95.93	95.96	96.01	**96.19**	96.14	96.00
Params(K)		**115.64**	128.44	144.44	163.65	186.04	211.65
FLOPs(M)		**151.20**	167.79	188.53	213.41	242.44	275.62
OA%	UP	85.37	86.24	86.17	**86.64**	86.51	86.60
AA%		88.32	88.25	88.71	**88.87**	88.53	88.76
Kappa×100		81.23	82.29	82.27	**82.82**	82.63	82.71
Params(K)		**115.00**	127.80	143.80	163.00	185.40	211.00
FLOPs(M)		**151.19**	167.78	188.52	213.40	242.43	275.61
OA%	PC	97.41	97.50	97.36	**97.75**	97.23	97.39
AA%		94.12	94.58	94.48	**94.99**	94.13	94.51
Kappa×100		96.34	96.47	96.27	**96.82**	96.09	96.31
Params(K)		**115.00**	127.80	143.80	163.00	185.40	211.00
FLOPs(M)		**151.19**	167.78	188.52	213.40	242.43	275.61

respectively, and analyze the OA, AA, and Kappa in the target domain dataset. As can be seen from Table VII, $\lambda = 0.1$ produces the best results, which is more suitable for the adopted HSIs.

The size of the convolutional kernel is also an important parameter in ADLKAN. The larger the convolutional kernel, the larger the perceptual field of view and the more information extracted. However, as the convolutionAL kernel increases, the model parameters and training costs increase rapidly. Therefore, we set the convolutional kernel size to $7 \times 7, 9 \times 9, 11 \times 11, 13 \times 13, 15 \times 15$, and 17×17 to select the appropriate size for large-kernel convolution in LKA. We analyzed OA, AA, the Kappa, and the number of parameters and floating point operations (FLOPs) on the target domain dataset. From Table VIII, we can see that 13×13 achieves the best performance and is more suitable for our HSI classification.

D. Classification Results of Different Methods

To verify the effectiveness of the proposed method in the few-shot classification task, ADLKAN was compared with nine methods, including the traditional method of SVM [8], deep learning methods 3D-CNN [16], SSRN [17] and SDRN [61], attention-based method SSLKA [28], and cross-domain small sample learning methods CMFSL [39], DCFSL [37], ADAFSL [38], FDFSL [62], and MLPA [63].

For noncross-domain methods, five labeled samples from each category in the target domain were randomly selected to construct the training set and the remaining target domain samples were used as the testing set. For DCFSL, CMFSL, ADAFSL, FDFSL, MLPA, and our proposed method, the Chikusei dataset was used as the source domain dataset, and five labeled samples were randomly selected from each category as the supervision samples of the target domain. To eliminate the randomness of random selection, we conducted ten experiments for each method and calculated OA, AA, and Kappa to measure the performance of different methods.

1) Results on the IP Dataset: Table IX lists the results of our proposed method and other methods on the IP dataset, where the best results are shown in bold. On the IP dataset, compared with the best deep learning method SSLKA, our method improves OA by 5.25%, AA by 2.73%, and Kappa by 5.8%. Compared with the best cross-domain method FDFSL, our OA improves by 7.54%, AA improves by 5.94%, and Kappa improves by 8.45%. In addition, our method shows better classification accuracy in categories, such as Alfalfa, Cron-notill, Cron-mintill, Corn, Grass-pasture, and Woods. Among them, OA improves the most in the Cron-notill category, compared with SSLKA, our method improves OA by 18.63%, and compared with FDFSL, it improves by 18.28%. In Fig. 12, it can be seen that these categories have smoother classification effects than other methods. This proves the effectiveness of the proposed ADLKAN.

2) Results on the Salinas Dataset: Table X lists the results on the Salinas dataset, where the best results are shown in bold. On the Salinas dataset, our method improves OA by 2.88%, AA by 1.27%, and Kappa by 3.22% compared with the best deep learning method SSLKA. Compared with the best cross-domain method FDFSL, OA improves by 1.61%, AA improves by 1.65%, and Kappa by 1.78%. It shows better classification results in the smooth follow categories, Grapes not trained, Soil vinyard develop, Corn senesced green weeds, Vinyard not trained, Lettuce romaine 7 wk, and Vinyard vertical trellis categories. Among them, OA improves the most in the Vinyard vertical trellis category, which is 16.58% higher than SSLKA and 2.50% higher than FDFSL. As can be seen from Fig. 13, these land cover types have smoother classification results than other methods.

3) Results on KSC Dataset: Table XI lists the results on the KSC dataset. Due to the effective feature extraction of the adversarial cooperative spectral–spatial attention block and the effective mitigation of domain shift by the multiscale conditional domain discriminator, our method also shows better performance than others. Compared with the best deep learning method SSLKA, OA is improved by 2.96%, AA is improved by 3.44%, and the Kappa is improved by 3.3%. Compared with the best cross-domain method FDFSL, OA is improved by 3.52%, AA is improved by 4.27%, and the Kappa

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TABLE IX

CLASSIFICATION RESULT (%) FOR THE IP DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM [8]	3D-CNN [16]	SSRN [17]	SDRN [61]	SSLKA [28]	DCFSL [37]	CMFSL [39]	ADAFSL [38]	FDFSL [62]	MLPA [63]	ADLKAN
1	93.90±9.20	72.11±18.09	76.05±18.25	96.58±6.23	99.21±1.21	92.68±9.82	98.05±5.09	97.80±2.30	97.07±3.24	97.56±1.54	**99.76±0.73**
2	17.43±10.91	25.95±11.98	22.75±6.58	45.08±11.40	50.70±11.64	43.62±10.79	43.72±10.97	55.29±12.59	51.05±7.38	47.72±10.99	**69.33±11.86**
3	18.75±18.30	17.59±9.84	25.15±9.51	45.63±9.09	65.08±10.20	48.68±7.20	57.81±6.60	59.03±7.56	65.28±5.99	49.44±7.48	**67.12±8.11**
4	36.81±9.96	33.23±7.44	30.49±11.21	68.92±12.24	86.77±14.62	79.18±11.93	84.66±12.82	84.48±9.77	82.37±13.30	82.54±11.44	**91.64±8.51**
5	27.95±21.75	46.37±19.40	50.17±11.54	74.72±9.39	78.46±8.61	71.97±8.40	76.28±8.95	79.25±8.39	71.55±11.68	72.74±7.30	**80.40±7.43**
6	40.61±7.30	48.72±18.62	73.70±9.72	87.04±11.07	**94.38±4.16**	86.32±5.43	82.30±7.21	85.37±9.79	87.56±5.59	85.72±5.49	93.88±2.98
7	92.61±1.99	76.11±19.41	95.00±5.80	100.00±0.00	100.00±0.00	97.83±4.01	100.00±0.00	99.13±1.74	99.57±1.30	98.70±3.91	100.00±0.00
8	32.62±15.98	80.84±1080	58.89±13.91	71.56±11.67	**95.42±9.25**	83.38±13.03	87.00±12.46	90.61±11.10	87.38±11.45	86.09±13.60	94.23±9.81
9	84.67±11.18	97.27±4.17	99.09±2.73	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	98.67±4.00	100.00±0.00
10	35.69±25.74	26.20±20.21	47.39±14.60	64.99±10.13	**77.32±4.74**	60.78±10.26	64.19±7.12	64.81±6.18	61.77±13.04	67.14±7.32	76.07±7.38
11	37.76±23.55	**69.02±10.79**	52.93±17.95	48.90±19.43	61.17±12.38	58.56±12.45	60.32±12.67	60.98±11.90	63.92±6.31	60.32±11.84	67.42±10.16
12	22.37±12.84	28.08±13.27	21.90±12.21	51.74±8.69	64.13±14.66	46.55±10.12	48.13±11.41	56.77±10.41	58.50±10.32	48.25±9.55	**72.26±16.26**
13	94.55±2.68	89.06±10.14	90.47±10.39	99.11±0.74	**99.53±0.64**	99.15±1.16	99.45±0.69	97.40±3.32	98.00±2.07	99.00±1.16	98.45±1.59
14	69.51±12.97	69.14±13.35	78.46±8.34	84.32±8.35	87.33±8.36	84.23±9.96	79.17±11.90	86.55±8.88	88.71±3.97	88.71±6.29	**90.04±8.77**
15	13.39±9.16	32.62±13.06	44.37±20.37	72.21±15.97	81.78±12.89	67.09±9.82	71.92±9.82	75.43±11.26	79.50±12.99	73.18±12.17	**86.25±11.05**
16	83.41±6.00	82.98±10.14	97.50±2.58	99.88±0.36	**100.00±0.00**	99.20±1.61	99.66±0.73	97.05±4.07	97.73±2.49	98.98±2.18	98.07±3.77
OA%	36.68±4.12	48.86±3.12	49.42±3.98	62.25±4.47	72.45±3.85	64.36±2.59	65.75±2.64	69.69±3.62	70.16±2.42	67.11±2.00	**77.70±3.54**
AA%	49.50±1.40	55.96±2.74	60.2±1.35	75.67±2.39	83.83±1.45	76.20±1.11	78.29±1.14	80.62±2.01	80.62±1.91	78.42±1.34	**86.56±1.75**
Kappa×100	29.40±3.62	41.33±3.54	43.09±3.84	57.70±4.59	69.09±4.16	59.90±2.71	61.50±2.80	65.86±3.98	66.44±2.68	63.01±2.17	**74.89±3.88**

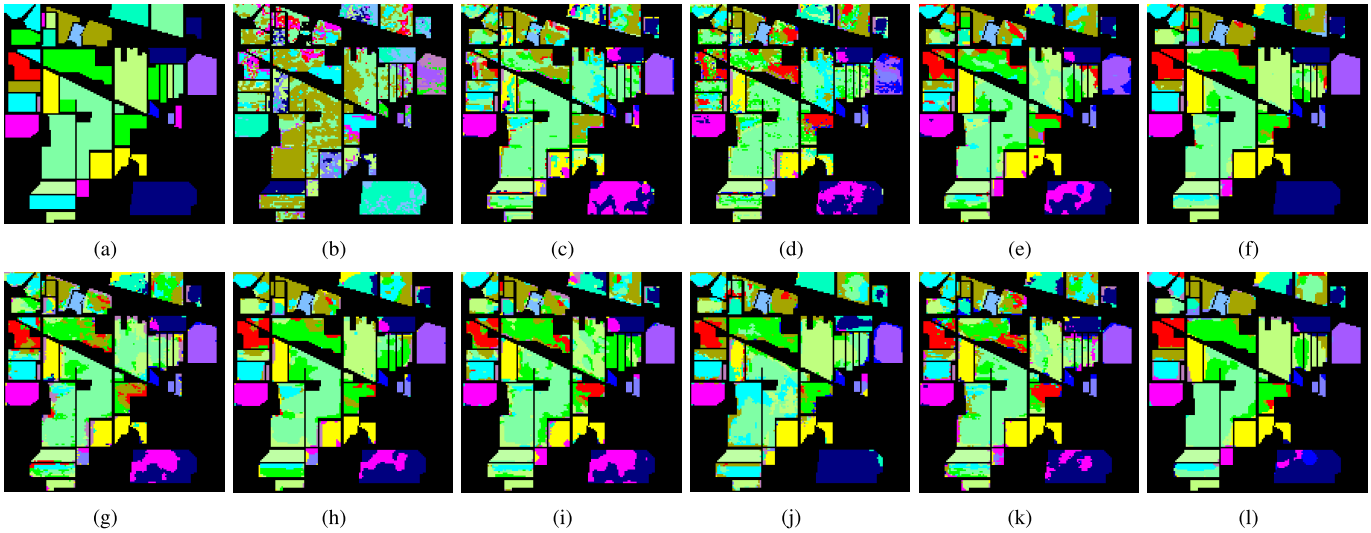


Fig. 12. Classification maps obtained via various algorithms for the IP dataset. (a) Ground-truth map, (b) SVM (36.68%), (c) 3D-CNN (48.58%), (d) SSRN (49.42%), (e) SDRN (62.25%), (f) SSLKA (72.45%), (g) DCFSL (64.36%), (h) CMFSL (65.75%), (i) ADAFSL (69.69%), (j) FDFSL (70.16%), (k) MLPA (67.11%), and (l) ADLKAN (77.70%).

TABLE X

CLASSIFICATION RESULT (%) FOR THE SALINAS DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM [8]	3D-CNN [16]	SSRN [17]	SDRN [61]	SSLKA [28]	DCFSL [37]	CMFSL [39]	ADAFSL [38]	FDFSL [62]	MLPA [63]	ADLKAN
1	99.19±0.48	66.74±35.79	90.89±7.07	**99.61±0.47**	99.32±1.29	99.05±1.04	98.93±1.35	99.14±0.89	98.77±0.98	99.01±1.37	99.20±1.33
2	62.22±11.21	99.17±0.51	98.01±1.07	99.67±0.07	**100.00±0.00**	99.23±0.93	99.46±0.53	99.53±0.56	99.17±2.03	99.51±0.64	99.84±0.21
3	36.67±17.31	89.49±11.33	80.44±10.83	**95.94±5.36**	95.44±7.04	91.81±10.61	90.40±10.43	91.86±11.38	88.65±12.99	88.98±12.52	93.92±10.72
4	97.19±0.62	97.75±2.96	98.26±1.07	98.95±1.19	99.28±1.19	99.41±0.59	**99.54±0.45**	98.29 ±1.68	99.42±0.33	99.12±0.63	99.16±0.97
5	96.43±3.99	90.58±6.63	89.65±5.84	93.68±2.70	94.12±3.97	91.51±3.76	92.96±2.44	94.75±3.60	92.79±4.01	91.18±5.39	**95.10±4.22**
6	96.94±1.02	99.22±0.76	97.80±2.31	99.80±0.26	**99.97±0.06**	99.23±0.97	99.35±1.12	97.56±2.37	99.25±0.95	99.31±0.94	99.21±1.09
7	98.39±1.58	97.77±3.42	94.62±1.96	99.89±0.25	**99.94±0.09**	99.48±0.58	99.25±1.01	94.25±15.64	99.43±0.51	99.21±1.48	99.65±0.62
8	61.62±23.64	60.21±14.11	58.90±28.45	62.21±9.43	79.16±4.70	74.95±9.66	74.40±14.16	75.18±11.22	79.37±5.15	76.97±7.59	**81.18±8.93**
9	93.48±6.11	94.54±3.15	94.90±4.96	97.02±2.33	98.74±1.31	98.92±1.28	98.92±1.89	99.37±0.89	98.37±2.34	99.39±0.80	**99.85±0.27**
10	16.39±22.55	72.23±22.40	65.97±14.99	88.11±6.98	88.92±6.60	84.85±8.75	84.52±7.93	86.90±6.29	84.55±7.48	85.33±6.68	**91.43±3.66**
11	75.32±22.01	93.79±3.13	91.50±4.39	97.25±2.24	99.28±0.72	97.45±2.02	98.61±1.24	98.17±1.27	95.77±5.82	98.17±1.30	**99.35±0.78**
12	90.32±5.32	98.58±2.34	96.65±2.84	99.21±1.16	**99.77±0.24**	99.39±0.78	99.69±0.36	99.55±0.40	99.81±0.20	99.53±0.52	99.64±0.25
13	98.70±0.47	98.55±1.42	99.37±0.96	98.27±1.60	**99.50±0.78**	99.45±0.62	98.73±1.74	99.13±1.59	98.31±1.51	98.87±0.74	99.13±1.16
14	88.89±1.47	94.79±2.34	92.23±3.98	98.22±1.32	99.23±1.41	98.12±1.09	98.87±0.86	98.56±1.55	99.53±0.36	99.27±0.80	**99.73±0.30**
15	49.82±18.88	55.15±15.15	46.26±27.89	69.35±7.92	66.87±16.25	76.49±4.94	75.35±9.14	77.07±5.97	82.64±7.21	74.58±6.29	**83.45±7.55**
16	59.96±3.18	79.91±12.46	81.72±6.09	93.77±3.95	95.49±3.87	93.32±4.01	92.22±4.56	89.01±6.42	93.10±4.51	91.53±5.67	**95.60±4.38**
OA%	71.53±3.18	79.93±3.19	78.18±2.19	86.00±1.38	89.64±2.15	89.32±2.13	89.06±2.26	89.19±3.05	90.91±1.36	89.42±1.64	**92.52±1.60**
AA%	76.35±1.44	86.78±3.70	86.07±1.22	93.20±0.55	94.69±1.25	93.92±1.25	93.82±0.98	93.65±1.49	94.31±1.02	93.75±1.23	**95.96±0.95**
Kappa×100	68.39±3.36	77.71±3.48	75.77±2.28	84.47±1.52	88.46±2.41	88.15±2.34	87.86±2.47	88.00±3.35	89.90±1.50	88.25±1.80	**91.68±1.76**

is improved by 3.91%. Among them, Scrub, Willow swamp, Slash pine, Broadleaf, Cattail marsh, Salt marsh, and Mud flats show better classification performance. Compared with other methods, ADLKAN improves OA by 2%∼10% in these categories. It can be seen from Fig. 14 that our method has better classification effects in these categories.

4) Results on the UP Dataset: Table XII shows the results on the UP dataset, where the best results are shown in

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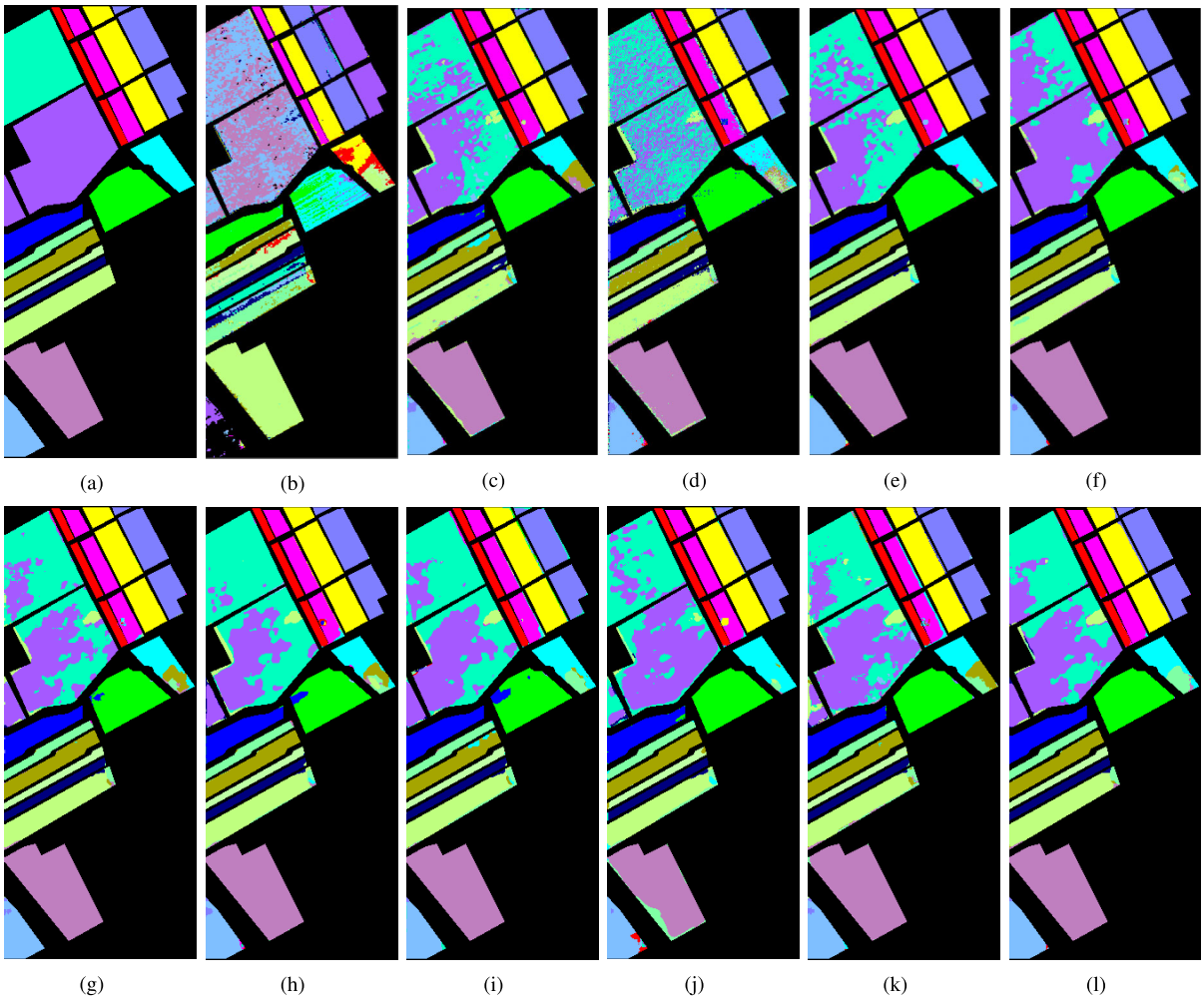


Fig. 13. Classification maps obtained via various algorithms for the Salinas dataset. (a) Ground-truth map, (b) SVM (71.53%), (c) 3D-CNN (79.93%), (d) SSRN (78.18%), (e) SDRN (86.00%), (f) SSLKA (89.64%), (g) DCFSL (89.32%), (h) CMFSL (89.06%), (i) ADAFSL (89.19%), (j) FDFSL (90.91%), (k) MLPA (89.42%), and (l) ADLKAN (92.52%).

TABLE XI

CLASSIFICATION RESULT (%) FOR THE KSC DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM [8]	3D-CNN [16]	SSRN [17]	SDRN [61]	SSLKA [28]	DCFSL [37]	CMFSL [39]	ADAFSL [38]	FDFSL [62]	MLPA [63]	ADLKAN
1	46.39±26.87	85.83±4.23	91.30±2.26	93.06±2.97	96.71±1.46	97.29±1.74	95.21±3.32	95.82±3.69	97.87±1.13	96.65±2.60	**98.75±1.51**
2	30.80±13.08	57.71±19.98	59.95±8.74	89.49±4.21	92.75±4.73	84.33±12.70	91.81±9.00	84.92±8.28	85.38±13.07	85.63±9.91	**95.63±7.18**
3	36.53±21.75	72.38±13.94	62.42±17.15	78.49±11.33	**95.71±3.50**	85.70±8.77	90.32±7.66	89.40±7.29	89.36±9.11	86.29±7.04	92.91±10.28
4	33.44±23.97	23.83±8.40	21.98±6.67	44.49±12.96	59.96±18.03	49.11±16.65	53.89±10.04	51.58±13.41	63.44±15.27	49.35±14.68	**76.92±11.99**
5	20.58±16.96	46.00±19.35	39.86±16.71	63.79±14.74	83.43±8.16	71.92±11.45	83.21±6.69	75.19±10.67	82.88±8.15	75.71±10.65	**87.63±8.21**
6	28.53±22.38	53.11±12.39	73.93±6.62	60.63±9.24	90.83±11.01	92.14±4.16	91.38±5.35	89.46±9.87	91.83±7.23	93.66±5.03	**98.71±1.73**
7	45.30±14.70	91.48±9.39	93.52±6.01	94.89±10.24	**99.20±1.35**	96.00±8.93	93.80±8.81	94.90±6.80	99.40±1.28	94.20±10.49	96.80±6.84
8	14.65±16.58	41.57±13.94	45.18±11.13	81.88±14.44	96.02±3.25	86.67±11.55	90.61±9.08	93.47±8.00	95.38±3.63	88.15±14.21	**97.70±3.55**
9	52.93±19.65	71.51±8.67	91.43±8.08	82.47±9.98	90.90±6.89	91.28±8.33	89.73±12.20	94.29±7.02	85.71±10.16	92.56±7.52	**95.22±7.19**
10	43.78±9.41	73.66±8.84	37.15±18.62	86.78±11.31	96.64±3.57	95.44±3.97	96.19±3.11	96.49±3.16	96.52±5.11	96.67±6.67	**99.52±0.47**
11	87.34±11.46	93.78±3.69	96.98±2.23	95.47±3.44	96.02±3.84	95.85±5.01	97.51±4.04	96.30±4.62	97.56±3.35	96.57±4.06	**98.09±2.11**
12	39.01±26.98	86.02±9.39	68.09±17.07	88.48±5.48	92.67±4.8	93.45±5.69	93.43±5.36	92.63±4.64	94.74±4.71	93.63±5.78	**97.87±2.63**
13	98.50±4.20	100.00±0.00	99.39±0.81	100.00±0.00	100.00±0.00	100.00±0.00	99.99±0.03	99.97±0.07	100.00±0.00	100.00±0.00	100.00±0.00
OA%	52.73±2.78	75.57±3.57	74.54±2.73	86.14±0.98	93.62±1.51	91.17±1.18	92.29±1.58	92.14±1.55	93.06±1.50	91.74±1.12	**96.58±1.20**
AA%	44.45±1.95	68.99±4.39	67.78±3.22	81.53±1.75	91.60±2.20	87.63±1.58	89.78±1.20	88.80±2.22	90.77±1.79	88.39±1.42	**95.04±1.38**
Kappa×100	47.43±3.01	72.79±3.97	67.78±3.23	84.56±1.10	92.89±1.68	90.17±1.32	91.42±1.75	91.25±1.72	92.28±1.67	90.81±1.25	**96.19±1.33**

bold. Our method improves OA by 3.56%, AA by 3.41%, and Kappa by 4.79% compared with the FDFSL method. Compared with the best deep learning SDRN, OA improves by 3.3%, AA by 0.19%, and Kappa by 3.95%. It shows better classification results in Asphalt, Meadows, Cravel, Bitumen, and Bricks categories. This proves the effectiveness of the proposed ADLKAN. Among them, the Bricks category OA has the largest improvement. Compared with SDRN, it has increased by 14.31%, and compared with FDFSL, it has increased by 2.87%. This demonstrates the effectiveness of the proposed ADLKAN. As can be seen from Fig. 15, the map generated by ADLKAN is most similar to the ground-truth map compared with other methods.

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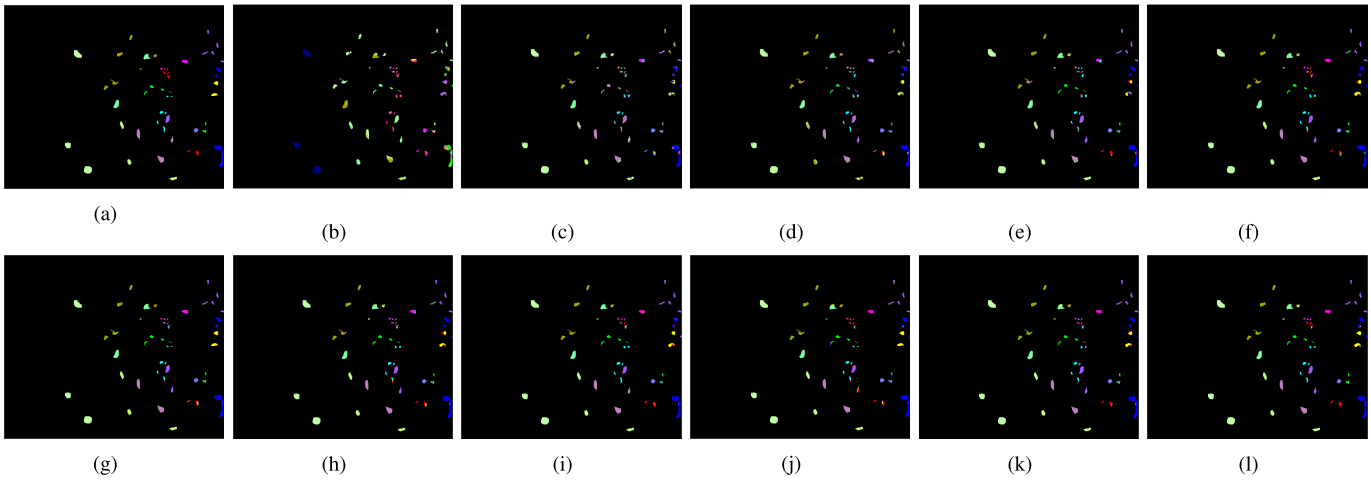


Fig. 14. Classification maps obtained via various algorithms for the KSC dataset. (a) Ground-truth map, (b) SVM (52.73%), (c) 3D-CNN (75.57%), (d) SSRN (74.54%), (e) SDRN (86.14%), (f) SSLKA (93.62%), (g) DCFSL (91.17%), (h) CMFSL (92.29%), (i) ADAFSL (92.14%), (j) FDFSL (93.06%), (k) MLPA (91.74%), and (l) ADLKAN (96.58%).

TABLE XII
CLASSIFICATION RESULT (%) FOR THE UP DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM [8]	3D-CNN [16]	SSRN [17]	SDRN [61]	SSLKA [28]	DCFSL [37]	CMFSL [39]	ADAFSL [38]	FDFSL [62]	MLPA [63]	ADLKAN
1	58.09±9.24	56.09±10.37	69.49±17.58	87.58±5.79	88.29±6.97	78.05±9.20	79.94±7.37	86.78±4.10	85.83±5.76	79.21±10.43	**90.49±3.51**
2	43.74±28.89	77.44±9.92	78.76±9.39	77.91±9.10	70.62±10.08	83.30±8.97	82.54±11.17	82.83±10.10	83.14±8.12	83.81±7.19	**84.39±10.58**
3	28.14±14.65	56.53±10.91	63.04±23.52	82.32±11.28	72.61±12.78	63.78±11.10	64.87±10.61	66.82±13.06	67.22±5.01	67.78±9.73	**72.62±8.34**
4	88.04±4.63	69.45±12.10	83.81±11.02	93.02±3.93	**93.09±5.54**	92.49±4.36	92.79±5.89	87.37±5.79	91.82±3.81	95.06±2.00	91.52±4.26
5	94.19±6.30	94.47±4.58	97.79±2.14	100.00±0.00	**99.99±0.02**	99.55±0.47	99.10±1.66	98.66±1.33	99.40±0.87	99.19±1.31	99.53±0.51
6	43.16±24.99	38.85±17.08	70.06±25.41	**88.59±7.39**	68.05±15.68	74.20±8.56	74.49±15.51	75.25±11.87	69.98±12.91	76.55±8.96	84.80±10.02
7	91.58±5.50	85.37±5.10	61.42±33.32	94.72±3.47	89.37±8.27	79.26±8.87	79.43±7.36	85.44±5.54	89.71±7.76	77.19±9.46	**91.54±4.84**
8	59.49±19.49	49.42±70.71	49.62±25.50	74.29±11.67	88.41±4.38	65.07±12.94	81.36±11.87	76.06±14.85	85.73±7.66	67.63±13.76	**88.60±7.20**
9	**99.92±0.05**	97.41±1.62	97.49±2.85	99.73±0.43	98.71±1.77	99.08±0.78	97.53±3.45	98.94±1.86	96.30±2.65	98.91±1.25	96.33±2.90
OA%	53.96±10.70	66.80±4.30	73.85±3.97	83.34±4.67	78.41±3.06	80.28±3.54	81.71±4.78	82.44±4.35	83.08±2.75	81.29±2.89	**86.64±4.29**
AA%	67.37±1.32	69.47±3.53	74.61±5.69	88.68±2.11	85.46±2.79	81.64±1.81	83.56±2.53	84.25±2.21	85.46±2.04	82.37±1.79	**88.87±2.17**
Kappa×100	45.22±9.36	57.31±4.82	66.37±5.00	78.87±5.59	72.63±3.38	74.59±3.98	76.47±5.49	77.30±5.05	78.03±3.21	75.89±3.30	**82.82±5.08**

TABLE XIII
CLASSIFICATION RESULT (%) FOR THE PC DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM [8]	3D-CNN [16]	SSRN [17]	SDRN [61]	SSLKA [28]	DCFSL [37]	CMFSL [39]	ADAFSL [38]	FDFSL [62]	MLPA [63]	ADLKAN
1	99.54±0.11	99.10±0.47	99.33±0.26	99.72±0.29	99.49±0.24	99.69±0.28	99.71±0.22	99.77±0.17	99.53±0.27	99.65±0.27	**99.80±0.23**
2	69.96±12.45	94.41±2.45	93.94±2.42	**94.79±1.81**	91.12±6.07	91.26±3.50	91.06±3.13	91.82±2.66	91.80±3.58	91.87±3.58	92.85±3.10
3	69.03±19.56	69.04±28.97	80.65±21.09	87.48±10.13	83.94±17.53	93.61±2.21	**94.48±3.73**	92.44±4.00	92.47±7.10	94.34±2.84	93.71±3.55
4	35.21±21.26	91.89±5.05	79.60±18.45	93.24±9.02	94.24±5.79	92.36±7.14	91.53±8.57	93.57±8.58	92.79±6.27	91.87±6.98	**95.28±5.12**
5	42.70±11.96	74.65±15.44	79.77±13.66	89.42±6.74	75.65±13.85	91.40±5.18	90.30±5.38	91.69±5.34	**91.69±5.04**	91.55±5.98	90.86±5.80
6	93.84±6.97	76.24±10.10	96.15±2.83	**99.08±0.79**	97.11±1.96	95.85±3.95	96.60±3.61	98.24±0.85	98.12±1.12	96.23±3.01	98.32±0.85
7	40.91±26.51	80.50±3.99	77.66±9.74	85.15±3.74	83.72±3.70	84.61±1.77	86.37±3.37	85.69±2.51	87.49±4.77	84.49±2.63	**87.64±4.17**
8	65.30±8.56	94.56±3.49	97.15±2.01	**98.75±0.64**	97.39±1.97	98.63±0.49	98.61±0.72	98.66±0.91	98.99±0.75	98.46±0.99	98.55±1.60
9	98.63±1.31	99.97±0.06	97.96±2.64	**99.99±0.04**	98.98±2.22	99.21±0.77	99.45±0.65	99.62±0.37	98.18±1.01	98.83±1.61	97.86±1.33
OA%	80.54±2.41	93.39±1.89	95.52±1.29	97.61±0.44	96.05±1.03	97.33±0.35	97.42±0.35	97.63±0.44	97.66±0.37	97.32±0.61	**97.75±0.67**
AA%	68.35±2.92	86.71±4.68	89.13±4.37	94.18±1.61	91.29±2.42	94.07±0.71	94.24±1.03	94.61±0.80	94.56±0.70	94.14±1.08	**94.99±0.74**
Kappa × 100	73.16±3.11	90.66±2.65	93.66±1.86	96.62±0.62	94.41±1.46	96.23±0.49	96.36±0.50	96.45±0.62	96.69±0.53	96.22±0.85	**96.82±0.94**

5) Results on the Center of Pavia Dataset: Table XIII shows the results on the PC dataset, where the best results are shown in bold. On the Pavia Center Dataset, our method improves OA by 0.14%, AA by 0.81%, and Kappa by 0.2% compared with the best deep learning SDRN. Compared with the best cross-domain method FDFSL, OA is improved by 0.09%, AA is improved by 0.43%, and Kappa is improved by 0.13%. It shows the best classification results in the Water, Bricks, and Bare Soil feature categories. Among them, Bricks and Bare Soil all improve OA by about 1% compared with SDRN and FDFSL. As can be seen from Fig. 16, the map generated by ADLKAN is most similar to the ground-truth map compared with other methods.

E. Impact of Different Numbers of Labeled Samples

To analyze the effects of different numbers of labeled training samples on the compared methods, we randomly selected 1–10 labeled samples from each class in the target domain to construct a few-shot dataset. Then, the model was trained using the few-shot dataset. To save training time, a convolutional kernel of size $K = 7$ was used in ADLKAN for convolutional operations. As shown in Fig. 17, on the four datasets, the classification performance of various classification methods increases with the increase of training sample data, which is in line with the expectation. Compared with other methods, the proposed ADLKAN shows higher classification accuracy on datasets with different data volumes, demonstrating

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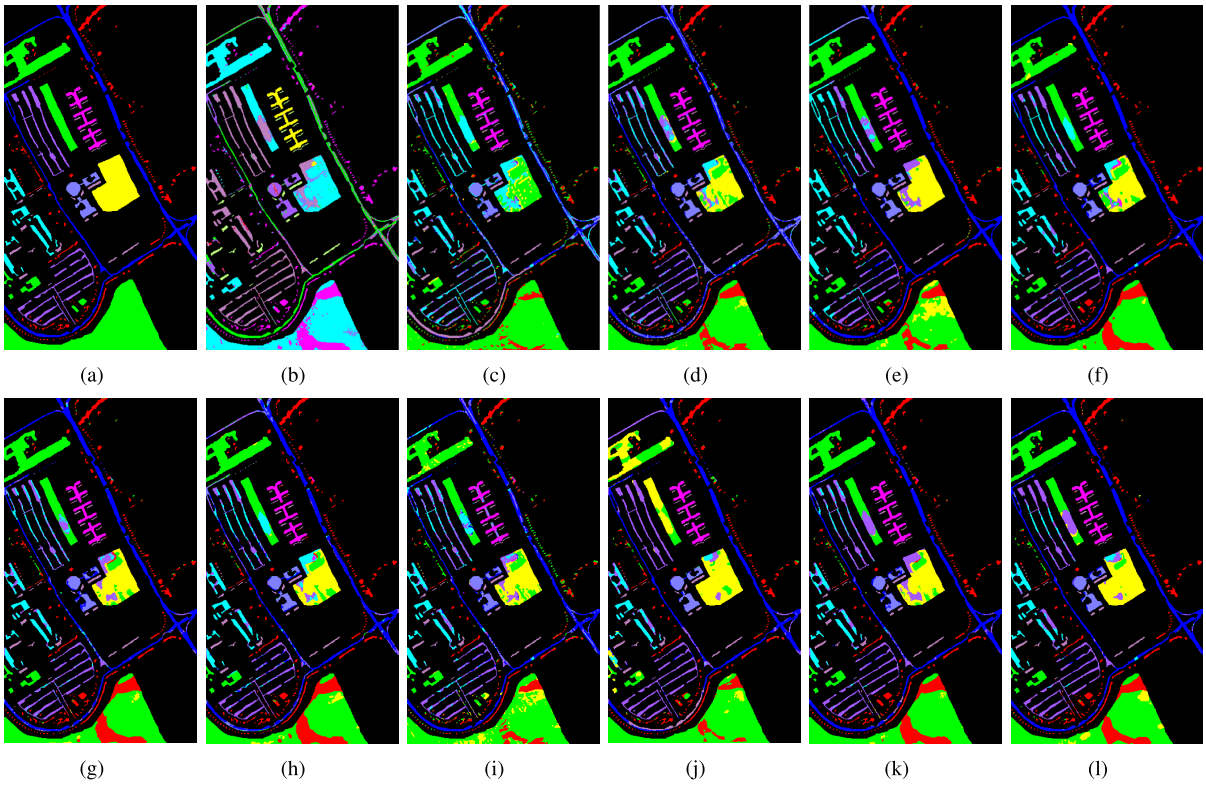


Fig. 15. Classification maps obtained via various algorithms for the UP dataset. (a) Ground-truth map, (b) SVM (53.96%), (c) 3D-CNN (66.80%), (d) SSRN (73.85%), (e) SDRN (83.34%), (f) SSLKA (78.41%), (g) DCFSL (80.28%), (h) CMFSL (81.71%), (i) ADAFSL (82.44%), (j) FDFSL (83.08%) (k), MLPA (81.29%), and (l) ADLKAN (86.64%).

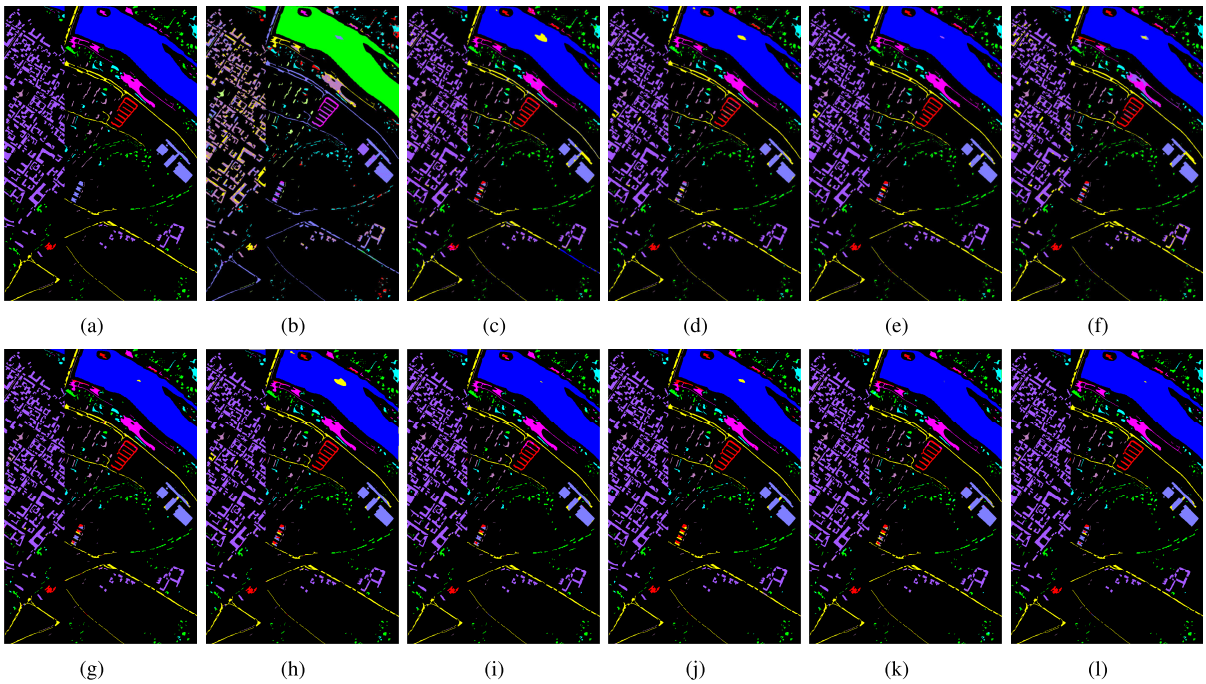


Fig. 16. Classification maps obtained via various algorithms for the PC dataset. (a) Ground-truth map, (b) SVM (80.54%), (c) 3D-CNN (93.39%), (d) SSRN (95.52%), (e) SDRN (97.61%), (f) SSLKA (96.05%), (g) DCFSL (97.33%), (h) CMFSL (97.42%), (i) ADAFSL (97.63%), (j) FDFSL (97.66%), (k) MLPA (97.32%), and (l) ADLKAN (97.75%).

its adaptability to changes in the number of labeled samples.

F. Ablation Studies

To further demonstrate the impact of each key component of our proposed ADLAKN, we conducted ablation experiments using the same experimental settings. The embedding feature extractor of ADLKAN consisted of a cooperative spectral–spatial AM with an LKA mechanism. We removed the AM, LKA, spatial attention branch, and spectral attention branch, i.e., (w/o) AM, (w/o) LKA, (w/o) Spa-Branch, and (w/o) Spe-Branch to test the corresponding

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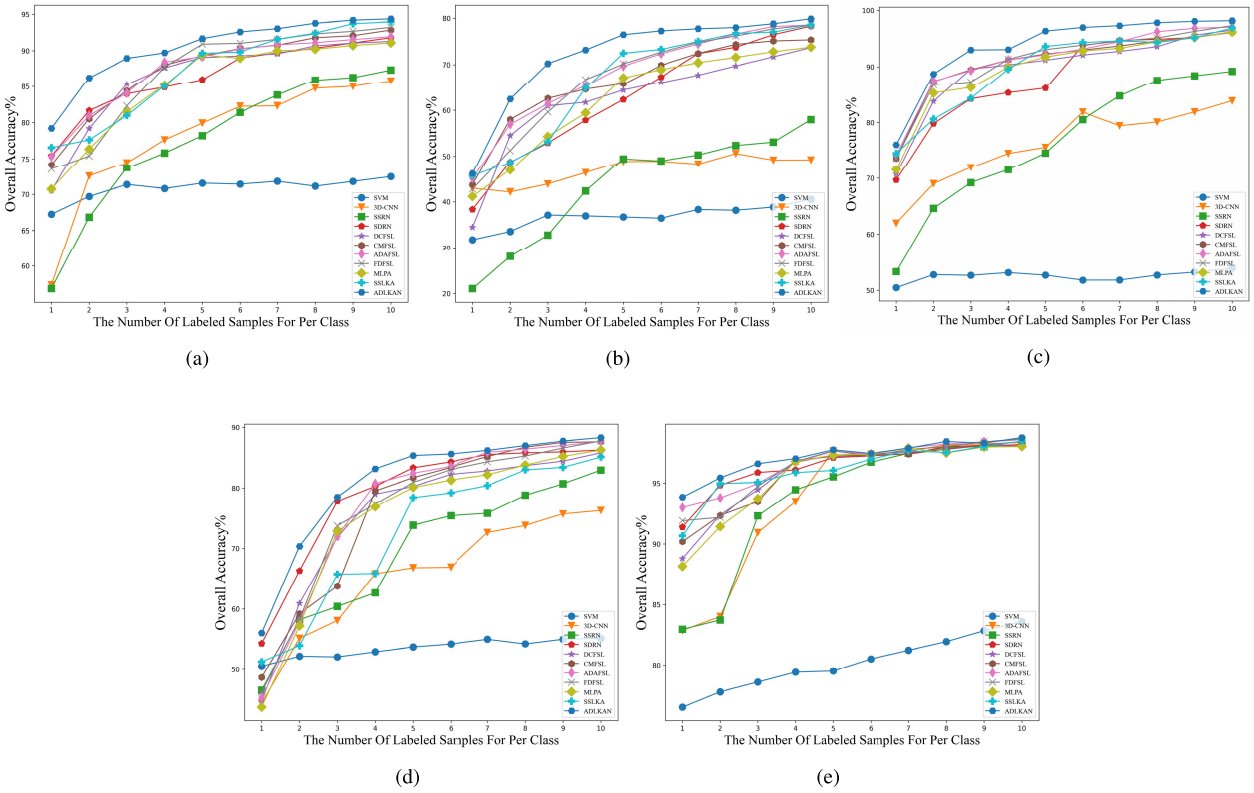


Fig. 17. Classification accuracy (OA%) with different numbers of labeled samples on target datasets. (a) Salinas. (b) IP. (c) KSC. (d) UP. (e) PC.

TABLE XIV
ABLATION STUDY OF DIFFERENT MODULES IN ADLKAN

Matrices	Datasets	w/o spa-Branch	w/o spe-Branch	w/o LKA	w/o AM	w/o Msc-discriminator	w/o adv-training	ADLKAN
OA%		72.25±4.00	74.96±3.87	70.87±3.88	67.41±2.91	75.10±3.14	75.83±3.64	**77.70±3.53**
AA%		85.15±2.17	85.12±2.17	82.54±2.22	79.36±1.55	84.90±1.67	85.67±2.06	**86.56±1.75**
Kappa×100	IP	72.15±4.41	71.85±4.26	67.25±4.24	63.36±3.11	71.98±71.99	72.83±4.02	**74.89±3.88**
Params(K)		**63.33**	103.70	99.73	94.67	164.13	164.13	164.13
FLOPs(M)		**83.04**	131.25	130.47	128.00	213.42	213.42	213.42
OA%		91.30±2.46	91.32±2.50	90.12±1.93	90.12±1.67	91.42±2.17	91.79±2.08	**92.52±1.60**
AA%		95.12±1.25	95.41±	94.76±1.09	94.77±0.92	95.09±1.23	95.48±1.19	**95.96±0.95**
Kappa×100	Salinas	90.33±2.70	90.37±2.75	89.02±2.12	89.03±1.84	90.47±2.38	90.89±2.28	**91.68±1.76**
Params(K)		**63.33**	103.70	99.73	100.21	164.13	164.13	164.13
FLOPs(M)		**83.04**	131.25	130.47	135.60	213.42	213.42	213.42
OA%		96.12±1.28	95.72±1.14	93.93±1.11	92.93±1.33	95.80±1.48	95.92±1.42	**96.58±1.20**
AA%		94.20±1.37	93.58±1.37	91.38±1.45	89.95±1.44	93.94±1.52	93.99±1.55	**95.04±1.38**
Kappa×100	KSC	84.82±1.42	95.24±1.27	93.24±1.23	92.13±1.48	95.32±1.64	95.45±1.58	**96.19±1.33**
Params(K)		**62.84**	103.21	99.25	98.17	163.65	163.65	163.65
FLOPs(M)		**83.03**	131.24	130.47	132.80	213.41	213.41	213.41
OA%		84.82±3.96	84.24±4.49	82.45±4.47	80.53±4.40	86.10±4.35	85.77±4.38	**86.64±4.29**
AA%		87.28±1.94	87.01±1.38	84.53±1.96	81.55±1.93	87.95±2.23	87.40±1.34	**88.87±2.17**
Kappa×100	UP	80.51±4.50	79.83±5.07	77.33±5.04	74.81±4.94	82.07±5.04	81.66±5.09	**82.82±5.08**
Params(K)		**62.20**	102.57	98.60	92.85	163.00	163.00	163.00
FLOPs(M)		**83.02**	131.23	130.46	125.50	213.40	213.40	213.40
OA%		97.19±0.56	97.48±0.55	97.18±0.50	97.59±0.82	97.54±0.44	97.47±0.42	**97.75±0.85**
AA%		93.85±1.09	94.16±0.82	93.85±0.87	94.51±0.76	94.21±0.41	93.80±0.61	**94.99±0.82**
Kappa×100	PC	96.03±0.79	96.44±0.77	96.02±0.69	96.60±1.34	96.52±0.62	96.42±0.56	**96.82±1.19**
Params(K)		**62.20**	102.57	98.60	92.85	163.00	163.00	163.00
FLOPs(M)		**83.02**	131.23	130.46	125.50	213.40	213.40	213.40

performance. To determine the impact of the multiscale conditional domain discriminator on the experimental results, we replaced it with the conditional domain discriminator in DCFSL [37] [(w/o) Msc-discriminator], and conducted experiments. Finally, we removed the adversarial training strategy of the embedding feature extractor [(w/o) adversarial training], to determine its impact on the classification effect. Table XIV shows the results of ablation studies conducted on IP, Salinas, KSC, UP, and PC datasets with OA, AA, and Kappa, respectively. The experimental results prove that the LKA mechanism, spatial attention branch, and spectral attention branch have a positive effect on enhancing the extraction of spectral–spatial features. The multiscale conditional domain discriminator and adversarial training strategy play an important role in our training framework. Therefore, each part in ADLKAN has an impact on the classification performance. Our proposed method can make full use of the global perception mechanism and channel adaptability of LKA

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TABLE XV
COMPUTATION TIME (SECONDS), FLOPS, AND PARAMETERS ON EACH DATASET WITH DIFFERENT METHODS

Datasets	Matrices	SVM [8]	3D-CNN [16]	SSRN [17]	SDRN [61]	SSLKA [28]	DCFSL [37]	CMFSL [39]	ADAFSL [38]	FDFSL [62]	MLPA [63]	ADLKAN
IP	Training Time	0.68	142.19	61.52	70.71	77.20	2095.03	1724.90	2848.33	2921.07	2155.34	1689.03
	Testing Time	0.20	11.27	14.01	2.49	0.52	1.56	0.75	1.57	2.11	1.50	1.43
	#params(K)	-	668.73	237.62	548.92	126.57	37.50	766.10	26.86	183.35	57.70	164.13
	FLOPs(M)	-	1883.50	798.91	11340.79	98.59	664.20	473.05	465.99	834.23	691.85	213.42
Salinas	Training Time	0.49	47.50	81.11	74.84	70.72	2030.28	1673.33	2826.60	2373.25	2172.51	1649.83
	Testing Time	0.83	0.72	1.65	14.40	2.80	5.35	3.91	6.68	13.24	5.64	4.85
	#params(K)	-	685.12	238.03	551.99	126.76	37.50	766.10	26.86	183.35	58.10	164.13
	FLOPs(M)	-	1929.12	799.23	11352.53	98.74	664.20	473.05	465.99	834.23	692.37	213.42
KSC	Training Time	0.41	56.50	72.92	64.20	72.14	2045	1512.30	2552.45	2174.96	2000.02	1637.28
	Testing Time	0.07	0.59	0.15	0.95	0.25	0.94	0.37	1.03	1.21	0.98	1.03
	#params(K)	-	469.82	235.07	552.16	125.24	37.01	766.10	26.38	183.35	54.81	163.65
	FLOPs(M)	-	1608.22	796.97	8874.30	97.66	664.19	473.05	378.61	834.23	688.74	213.41
UP	Training Time	0.34	61.25	55.24	69.50	59.03	1568.93	1219.75	2136.87	1836.86	1696.20	1396.16
	Testing Time	0.67	19.73	1.23	0.22	1.98	3.63	2.64	4.44	8.66	3.71	3.34
	#params(K)	-	167.48	227.45	540.83	121.46	36.37	766.10	25.74	183.35	46.87	163.00
	FLOPs(M)	-	775.73	791.07	854.09	94.86	664.18	473.05	262.11	834.23	679.26	213.40
PC	Training Time	0.02	40.62	52.01	27.17	33.52	1847.63	1219.75	1362.41	1828.58	1724.72	1379.67
	Testing Time	0.80	16.10	11.55	10.98	8.52	16.75	2.64	19.61	58.33	11.65	11.79
	#params(K)	-	165.18	213.47	268.09	121.41	36.37	766.10	25.74	183.35	46.77	163.00
	FLOPs(M)	-	764.35	766.87	424.42	94.82	664.18	473.05	262.11	834.23	679.14	213.40

to capture spectral–spatial features while easing to achieve better classification performance.

G. Computational Complexity

We analyze the training and testing time as well as the number of parameters and FLOPs to evaluate the computational complexity of the proposed methods. Table XV lists the training time, testing time, FLOPs, and parameters of all the above methods on the IP, Salinas, KSC, UP, and PC datasets (five labeled samples per class in the dataset). For the cross-domain methods (DCFSL, CMFSL, ADAFSL, FDFSL, MLPA, and our proposed ADLKAN), the training time is spent on transfer learning, and the testing time includes the time of fitting the NN classifier on five labeled target domain samples and the prediction time for the unlabeled samples of each HSI dataset. For the noncross-domain methods (SVM, 3D-CNN, SSRN, SDRN, and SSLKA), the training time is spent on model training with the target domain training set, and the testing time is spent on predicting the samples in the target domain testing set. Compared with other methods, the computation time, model parameters, and FLOPs of our proposed ADLKAN are all within an acceptable range.

V. CONCLUSION

In this article, we have proposed a novel method ADLKAN for cross-domain few-shot HSI classification. In contrast to the current cross-domain methods, ADLKAN can effectively extract global and local spectral–spatial features by multiple dilated small kernels and cooperative spectral–spatial attention blocks. Moreover, a multiscale domain discriminator is proposed to alleviate the domain shift problem. The proposed adversarial training method acting on both source domain FSL and target domain FSL can effectively improve classification accuracy. Extensive experiments on four benchmark datasets show that the proposed ADLKAN outperforms several state-of-the-art cross-domain few-shot HSI classification methods. In future work, we will further integrate LKA with domain adaptation to improve the classification performance.

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